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Transition towards decarbonised power systems and its socio-economic impacts in West Africa

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ABSTRACT

Pathways towards a defossilated sustainable power system for West Africa within the time horizon of 2015–2050 is researched, by applying linear optimisation modelling to determine the cost optimal generation mix to meet the demand based on assumed costs and technologies in 5-year intervals. Six scenarios were developed, which aimed at examining the impact of various policy constraints such as cross-border electricity trade and greenhouse gas emissions costs. Solar PV emerges as the prime source of West Africa's future power system, supplying about 81–85% of the demand in the Best Policy Scenarios for 2050. The resulting optimisation suggests that the costs of electricity could fall from 70 €/MWh in 2015 to 36 €/MWh in 2050 with interconnection, and to 41 €/MWh without interconnection in the Best Policy Scenarios by 2050. Whereas, the levelised cost of electricity without greenhouse emission costs in the Current Policy Scenario is 70 €/MWh. Results of the optimisation indicate that a fully renewables based power system is the least-cost, least-GHG emitting and most job-rich option for West Africa. This study is the first of its kind study for the West African power sector from a long-term perspective.

1. Introduction

Energy crisis and susceptibility to climate change are foreseen to constrain the future human and economic growth of West African (WA) countries [1]. Globally, the need for harmonised efforts to alleviate the danger of climate change and eradicate widespread energy poverty is apparent in the perspectives of the Paris Agreement on climate change and Sustainable Development Goal no 7 (SDG 7) [2]. In WA, a great deal of attention is directed towards the interrelated problems of energy crisis, climate change and energy security, which is characterised by growing demand, poor access to electricity and huge dependence on unsustainable biofuel [3]. In doing so, the Economic Community of West African States (ECOWAS) has adopted measure to streamline renewable energy (RE) and energy efficiency into their energy policies to tackle the predominant energy challenges in the region [4].

ECOWAS is comprised of 15 member states, and is characterised by diverse socio-economic, demographic and cultural backgrounds, each of this factor influence the region's electricity production and consumption [5]. ECOWAS with a growing population above 300 million, accounts for almost one-third of Sub-Saharan Africa's population, occupying an area of over five million square kilometres [6]. The regional gross domestic product (GDP) rebounded, averaging about 2.5% in 2017 from 0.5% in 2016, and is expected to increase to 3.9% in 2019 [7]. In spite of the region's abundant energy potential and progress achieved in the establishment of the regional power pool, the West African Power Pool (WAPP) [8], the region's electricity sector is underpowered with inadequate generation and transmission systems [9], leaving about 175 million people (48%) in the region unelectrified in 2016. Further, the total population relying on biomass for cooking was 263 million (75%) in 2015 [10]. Studies on energy consumption and economic growth nexus conclude that energy is a critical parameter for socio-economic development [11]. The socio-economic development of most WA countries is hampered by its underdeveloped energy sector [12]. Most ECOWAS countries rank among the poorest, having Low Human Development [5]. Access to electricity in the region is at 52%, with shortages of up to 80 h/month and yet electricity prices in WA remain among the costliest in the world, at 0.21 €/kWh, more than twice of the global average [13]. In 2016, the electrification rate was below 40% in 10 of the 15 countries, with Guinea-Bissau, Liberia, Niger and Sierra Leone occupying the

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Abbreviations

A-CAES	adiabatic compressed air energy storage
BPS(s)	Best Policy Scenario(s)
CAPEX	capital expenditures
CCGT	combined cycle gas turbine
CHP	combined heat and power
CPS(s)	Current Policy Scenario(s)
CSP	concentrating solar thermal power
ECOWASS	Economic Community of West African States
GT	gas turbine
GHG	greenhouse gas
HVDC	high voltage direct current
IRENA	International Renewable Energy Agency
IEA	International Energy Agency
LCOC	levelised cost of curtailment
LCOS	levelised cost of storage
LCOE	levelised cost of electricity
LCOT	levelised cost of transmission
OCGT	open cycle gas turbine
OPEX	operational expenditures
PHES	pumped hydro energy storage
PV	photovoltaic
RE	renewable energy
RoR	run-of-river
SNG	synthetic natural gas
SDGs	Sustainable Development Goals
ST	steam turbine
TES	thermal energy storage
VRE	variable renewable energy
WACC	weighted average cost of capital
WA	West Africa
WAPP	West Africa Power Pool

bottom: with 13%, 12%, 11% and 9% respectively [14]. The average annual electricity consumption in WA was about 145 kWh/capita in 2015 [15].

Nonetheless, the electricity supply gap is likely to increase as population, urbanisation and income are expected to rise, driving up electricity demand in the nearest future [9]. Electricity demand in the region is projected to increase fivefold by 2030, to 250 TWh, based on the ECOWAS Master plan [16]. The current grid capacity is not sufficient to cover the demand implying load shedding is becoming more prominent, driving consumers towards large-scale use of costly backup generation [17]. Due to significant under-capacity in electricity generation, some countries in the region such as Benin, Burkina Faso, Niger and Togo rely on electricity imports for a substantial share of their supply. As of 2015, the power generation capacity in the ECOWAS region is about 21 GW, producing about 57 TWh of electricity [15]. As shown in Fig. 1, more than half of the grid-connected capacities in the region are natural gas powered thermal plants, mostly in Nigeria, where it is the main power generation technology [18,19]. Nigeria, Ghana and Côte d'Ivoire account for more than 80% of installed capacity and generation [18]. In addition, the operating capacity is low in comparison to the overall existing capacity in most of the countries in the region.

In order to stem the energy crisis while contributing to the objectives of the Paris Agreement, the ECOWAS region aims to increase the share of grid-connected RE in the overall generation mix, including large hydropower, to 35% by 2020 and 48% by 2030. In addition, the share of rural population served by decentralised renewable energy systems is expected to reach 22% by 2020 and 25% by 2030 [3]. Over the period 1990–2015, CO_2 emissions in WA have increased by 68% reaching 139 Mt CO_{2eq} [20]. The electricity sector is the focus of many countries, in slowing down GHG emissions, as the sector accounts for one-third of the world's energy-related GHG emissions [21]. Therefore, a transition towards renewable electricity systems is essential, as the current systems dominated by fossil fuels are unsustainable on all accounts of social, economic and environmental criteria [22]. Over the past decades, the global trend in installed RE capacities has grown significantly from 995 GW in 2007–2179 GW at the end of 2017, and is dominated by solar photovoltaics (PV) and wind energy. In Africa, RE installed capacity grew from 23 GW in 2007 to 43 GW at the end of 2017. Much of this growth comes from solar PV (+98%), wind (+90%) and hydropower (+32%) [22]. RE has come to lead the new investments in the global power sector. As a result, the RE cost decline accelerates further, out-competing new built fossil capacities [23]. Solar PV utility-scale global levelised cost of electricity (LCOE) fell by 73% between 2010 and 2017 [24]. For instance, the differential between the LCOE for onshore wind and solar PV in South Africa is now 40% and almost 50% lower in price than newly built coal and nuclear, respectively [25].

The ECOWAS region has huge RE resource potential, widely distributed across the region and could provide low-cost and reliable energy supply [5]. Countless opportunities exist for deploying solar PV, wind energy, hydropower and biomass technologies across the region [5]. Currently, RE generation in WA is dominated by hydropower; and is even the main power source for some countries. Solar PV, wind energy and hydropower are anticipated to experience strong growth in the region's power mix [4]. However, these power sources in WA are governed by the monsoon, which causes seasonal variability [4]. Thus, the need for improved interconnections within the ECOWAS region is essential to achieve the synergetic effects by harnessing locally available RE resources, a solution to the challenges of low access and unreliable electricity supply in the region. Furthermore, the variability of RE supply may not be as significant, when viewed over larger geographic area [26]. Moreover, there is a spatio-temporal complementarity between various resources [27], which result in lower intermittency when examining the entirety of resources contribution, rather than a single resource contribution [26]. Additionally, energy system

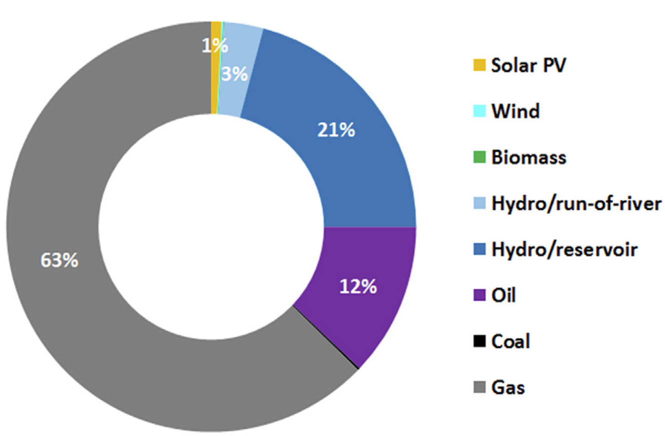

Fig. 1. West Africa power plant installed capacity by 2015 [18].

modelling is essential when assessing the least cost electricity expansion path for developing countries or regions [26]. Recent works have shown the possibility of achieving a fully renewable electricity system, as in the case of Nigeria [19], SSA [26] and global [28]. These studies have shown technical possibility and economic prospects of full defossilisation of the future power system, while considering all sustainability criteria. Both international agendas, the Paris Agreement and SDG 7 can be achieved by the deployment of RE technologies, in tackling the two major challenges of the 21st century; climate alteration and prevalent energy poverty [2]. Energy access is underlined in the SDG 7, while energy transition is highlighted in the Paris Agreement for mitigating climate change [2]. A brief review on RE share in the WA energy system is presented in Table 1.

So far, there is a lack of high temporal and spatial resolution energy transition studies for WA, which considers the impact of high penetration of RE in meeting the growing demand in the region under the operation of the WAPP electricity market. For these reasons, this study extends the investigation begun in Refs. [4,17,26,29–31], for a better understanding of the roles and benefits of flexible electricity generation, interconnected electricity networks and energy storage solutions in the transition towards a fully decarbonised power system and improved energy access for the 15 member states of ECOWAS, within the time horizon of 2015–2050. In addition, this study seeks to determine the least-cost and most-job enriching power system for WA. Furthermore, WA as an electricity island or individual country scenarios are compared towards a least-cost solution for the region. To this end, six scenarios were developed to fully understand the transition pathways for WA, under certain policy constraints, such as cross-border electricity trade and GHG emissions costs. These scenarios could cater, for instance, to policy decisions for decarbonising the WA electricity system within the time horizon of 2015–2050. The research investigates cost optimal generation mix to meet the demand based on assumed costs and technologies in 5-year intervals. The paper is organised as follows: section 2 describes the research methodology. Section 3 presents the result of the optimisation. Implications of the transition is discussed in section 4. Conclusions and policy measures are proposed in section 5.

2. Methodology

The ECOWAS power system optimisation was performed with the LUT Model described in Ref. [28,32]. The employment creation during the transition is analysed based on methodology presented in Ref. [33,34]. Fig. 2 illustrates the geographical scope of this study. West Africa was structured into 20 defined sub-regions based on the existing cooperation.

2.1. Model description

The LUT model a linear optimisation tool, which performs simulations for an entire year on an hourly resolution under certain operational conditions. The key function of optimisation algorithm is to minimise the system cost. To compute the lowest system cost, the model seeks to optimise the sum of installed capacities of each technology, operational expenditures, and costs of generation ramping. The WA power sector transition is simulated in 5-year time intervals under certain constraints. The model's main inputs and outputs are provided in Fig. 3. The model detail description can be found in Ref. [28].

In addition, the energy system takes into account electricity distributed generation by prosumers. The prosumer consumption is categorised as follows; residential, commercial and industrial. The prosumer sector is optimised exogenously on hourly resolution, the self-consumption model describes the optimal PV and storage (mainly battery) system size for the potential prosumers. The key function for self-consumers algorithm is to minimise cost of electricity consumed, estimated as the summation of prosumer generation, cost of grid electricity consumed and annual costs. Excess prosumer generation is sold to the grid at 0.02 €/kWh by prosumers, when their own demand is satisfied, but not more than 50% of total self-generation.

The model operates under two essential constraints:

1. The RE capacity share increase cannot exceed 4% per year (3% per year from 2015 to 2020), in order to avoid disruptions.
2. From 2015 onwards, no new conventional power plants would be installed, except gas turbines due to their lower carbon emissions, higher efficiency, and their ability to use sustainable synthetic biomethane and natural gas in the later phase. The current fossil-fueled plants are decommissioned based on their technical lifetimes [35].

2.2. Applied technologies

The ECOWAS power system is modelled with various technologies as show in Fig. 4, which include electricity generation, storage, and transmission. Existing transmission grid capacity was taken from Ref. [17], losses in transmission and distribution are considered during transition simulation [36]. Fig. 4 illustrates the LUT model components.

2.3. Technical and financial assumptions

The technical and financial assumptions introduced to the model are provided in the Supplementary Material (Tables S1–S4).

Table 1
Review on the share of renewable energy in West Africa.

Study	Year	Remark
IEA [17]	2014	The New Policies Scenario assumes RE installed capacity of 43 GW (38%) and fossil power plant of 70 GW (62%) by 2040. Gas dominates the installed capacity with 50 GW (44%), followed by hydropower with 24 GW (21%). Gas and hydropower contribute 249 TWh (53%) and 100 TWh (21%), respectively, of the total electricity generation at 474 TWh by 2040.
IRENA [29]	2015	RE installed capacity is 32 GW, and hydropower tops with about 22 GW by 2030 for the power sector. Hydropower dominates with 82 TWh for the total RE generation at 116 TWh (52%).
IRENA [30]	2018	This study reveals a growing share of RE in WA by 2030 under three scenarios. Depending on the scenario analysed, solar PV in the region ranges from 8.2 GW to 21.5 GW by 2030. While hydropower ranges from 11.4 GW to 11.5 GW, and wind energy remains constant at 1.6 GW for all scenarios. The remaining capacity is mainly supplied by gas in the range of 30 GW–36 GW by 2030.
Adeoye and Spataru [31]	2018	The study compares the baseline scenario of the 2025 power system with the renewable energy scenario of high penetration of solar PV. In the baseline scenario, gas-fired power plants contributes 55% of electric power generated, 27% from hydropower, 9% from coal and 8% from diesel. Under the renewable scenario, gas power plants account for 37% of electricity supply, 28% comes from solar PV, 25% from hydropower and 3% from diesel.

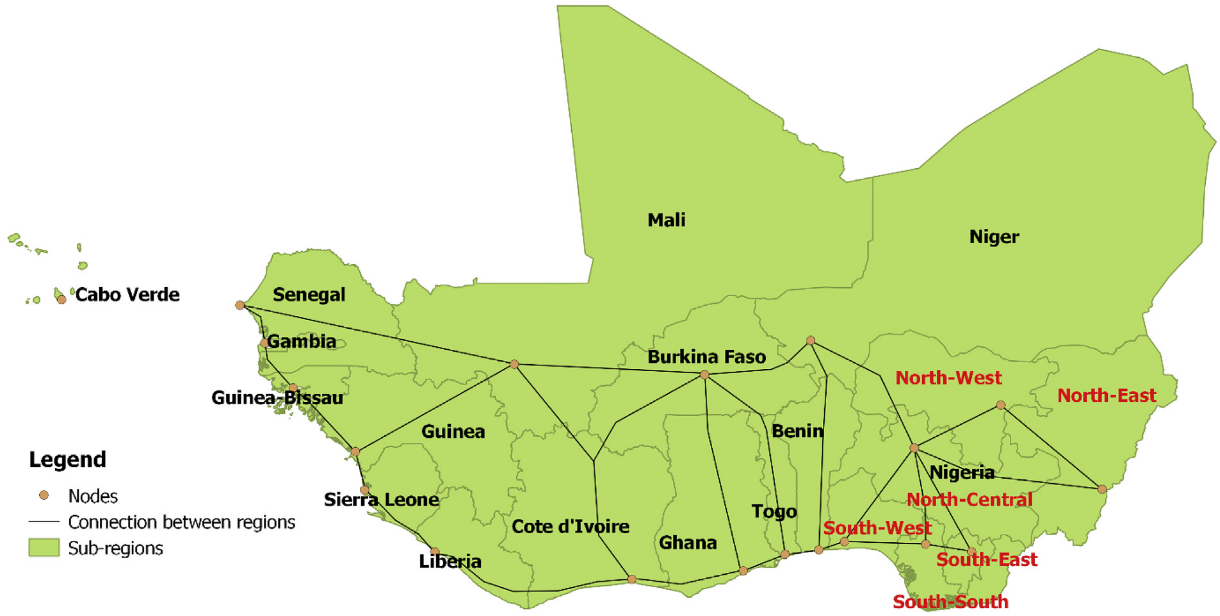

Fig. 2. The different sub-regions of West Africa.

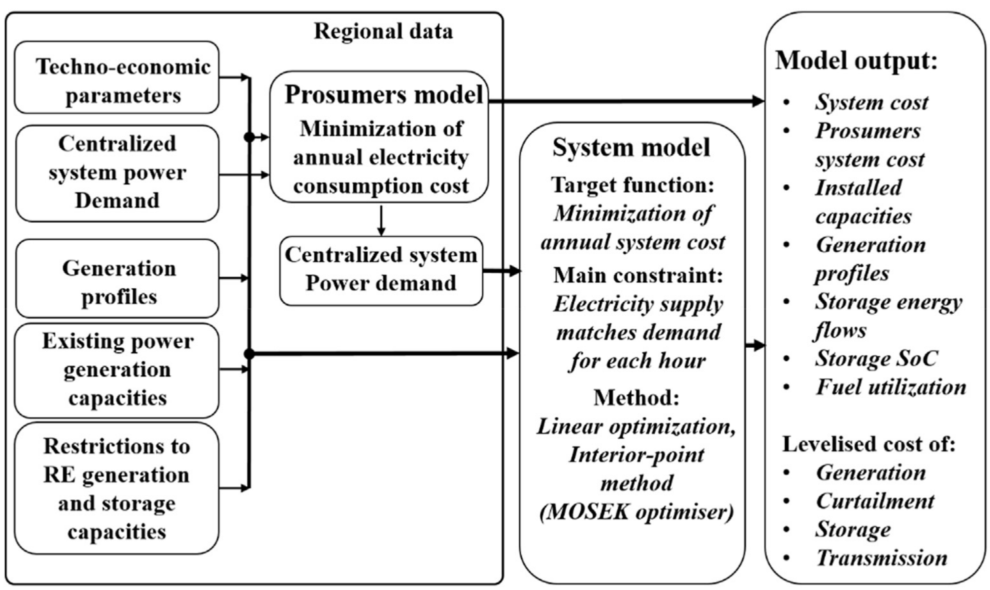

Fig. 3. Schematic of the LUT model [28].

In all the scenarios examined, a 7% weighted average cost of capital (WACC) was assumed, whereas 4% WACC is set residential PV self-consumption. A lower WACC is assumed for the PV self-consumption, as financial return expectations are lower. The residential, commercial and industrial consumers electricity prices were estimated till 2050 based on methodology described in Ref. [38,39]. The cost of electricity for each country in the region are provided in the Supplementary Material (Table S5).

The RE technologies upper limits were estimated based on methodology described in Ref. [32], existing installed capacities until 2015 are taken from Ref. [35] and set as lower limit. Absolute numbers of the upper and lower limit of all technologies are provided in the Supplementary Material (Tables S6 and S7). The required power capacities during transition for WA are provided in the Supplementary Material (Tables S8–S12).

2.4. Renewable resources potential

The wind energy, optimally tilted PV and solar CSP generation profiles, were estimated based on the methodology described in Ref. [32], and PV single-axis tracking according to Ref. [40], based on resource data of NASA [41,42] reprocessed by the German Aerospace Centre [43]. The hydropower generation profiles are estimated using daily resolved water flow data for the year 2005 [44]. The computed Full load hourly (FLH) for all resources can be found in the Supplementary Material (Tables S13–S17). Fig. 5 shows the annual FLH of various resources for entire WA. Biomass and waste potentials are provided in Ref. [45] and are classified according to Ref. [32]. Biomass costs are estimated using data from Ref. [46,47]. A gate fee of 50 €/ton was assumed for solid waste in 2015, which gradually increases up to 100 €/ton in 2050.

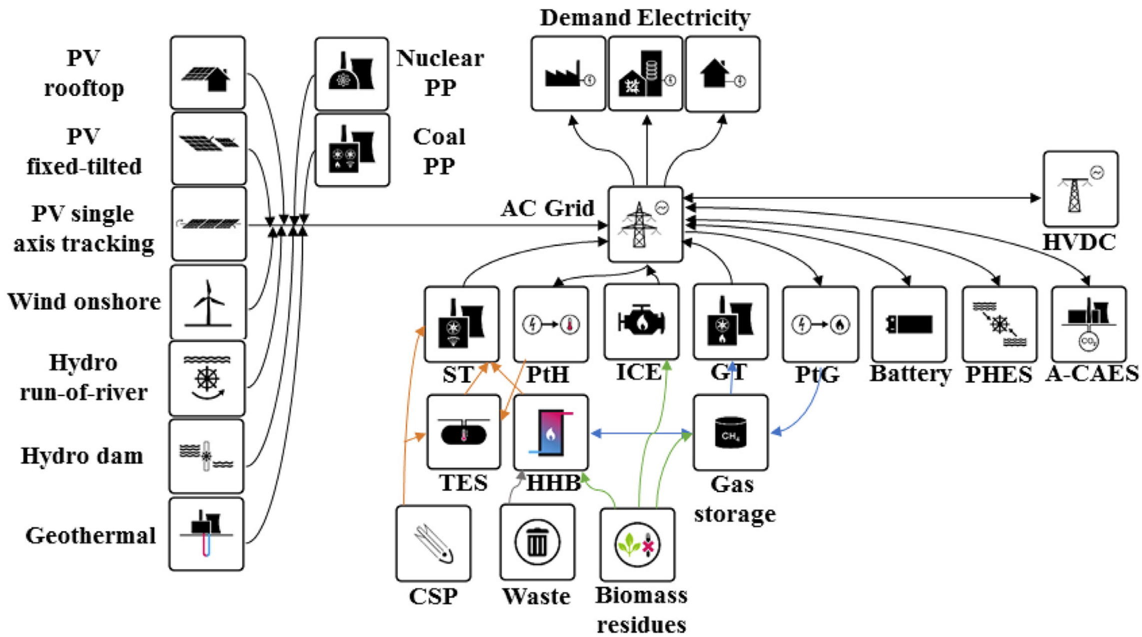

Fig. 4. Components of the LUT model [37]. Abbreviations not introduced elsewhere include PP - power plant, ST - steam turbines, PHES – pumped hydro energy storage, HHB – hot heat burner, ICE – internal combustion engine, PtH – power-to-heat, GT - gas turbines, PtG - power-to-gas, CSP – concentrated solar thermal power, A-CAES – adiabatic compressed air energy storage, TES – thermal-energy-storage.

2.5. Development of electricity demand

The WA electricity demand for the power sector is estimated to increase from 60 TWh in 2015 to about 667 TWh in 2050 [17], and absolute numbers are provided in the Supplementary Material (Table S5). Fig. 6 shows the aggregated load curve for WA. A regional compound average annual growth rate of about 7% in electricity demand drives the transition. The electricity demand is driven mainly by Nigeria, Ghana and Côte d'Ivoire, together they account for about 80% of the regional demand by 2050. The hourly load demand profiles are estimated based according to Refs. [48].

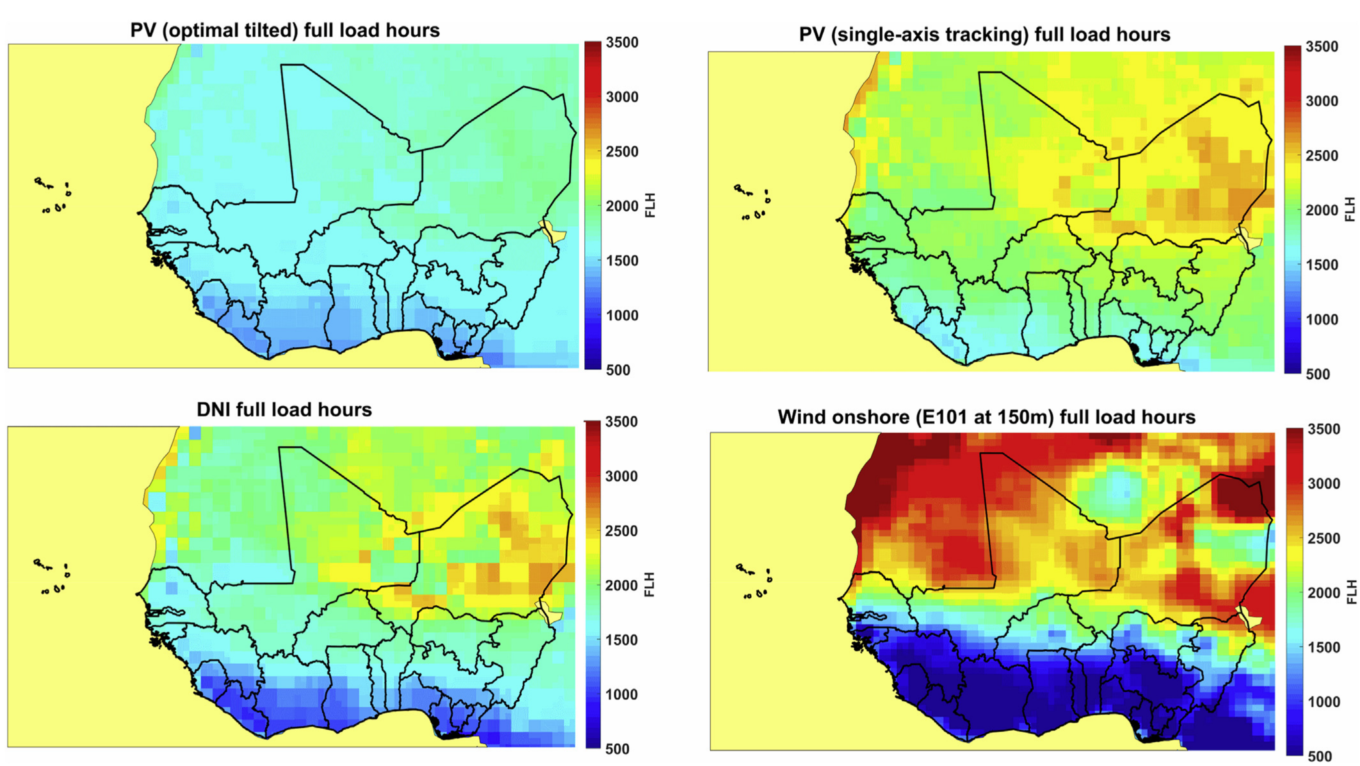

Fig. 5. Maps of West Africa showing annual full load hours for optimally tilted (top left) and PV single-axis tracking (top right), CSP solar field (bottom left), and wind (bottom right).

2.6. Scenario descriptions

In this study, six scenarios were studied for WA power system analyses, which are briefly described in Table 2.

3. Results

3.1. Electricity installed capacity and generation mix

Investments in power generation capacities are needed in WA, due to under-capacity power supply and rapid increase in electricity demand across the region. The installed capacities during the transition for all technologies under various scenarios are shown in Fig. 7 and absolute numbers are provided in the Supplementary Material (Tables S8–12). Table 3 presents the electricity installed capacity by technology for all scenarios in 2050. The respective power generation capacities in the BPSs is visualised first as shown in Fig. 7(a–d). Across the BPSs, solar PV dominates the installed capacities by 2050, with 284 GW in BPS-A, 284 GW in BPSnoCC-A, 328 GW in BPS-C, and 316 GW in BPSnoCC-C. Besides solar PV, a diverse technology mix can be observed in Fig. 7(a–d). Wind energy appears to be relevant in the BPSs across the area (BPS-A and BPSnoCC-A), while gas-based technologies contribution appears to be higher in the BPSs across the countries (BPS–C and BPSnoCC-C). The installed capacities in the CPSs are shown in Fig. 7(e–f). Gas turbines and hydropower dominates the installed capacities during the transition in the CPSs, followed by nuclear and coal, while the remaining capacities come from solar PV, wind energy and bioenergy. In 2050, gas turbine and hydropower installed capacities reach 72 GW and 28 GW, respectively, in the CPSs.

Fig. 8 shows the generation mix under various scenarios during the transition. Electricity generation increases for all scenarios to meet the demand over the transition in WA. In 2050, solar PV tops the generation mix in the BPSs, complemented by hydropower, bioenergy and wind energy as shown in Fig. 8(a–d). Solar PV contributes about 589 TWh in BPS-A, 590 TWh in BPSnoCC-A, 644 TWh in BPS-C and 619 TWh in BPSnoCC-C of the total electricity generation in 2050. Whereas, gas turbine dominates the power supply in the CPSs, followed by hydropower, coal and nuclear. By 2050, gas turbines account for 60% (407 TWh) of the generation and hydropower for 18% (120 TWh). Additional graphical results on the generation mix under various scenarios are available in the Supplementary Material (Fig. S1).

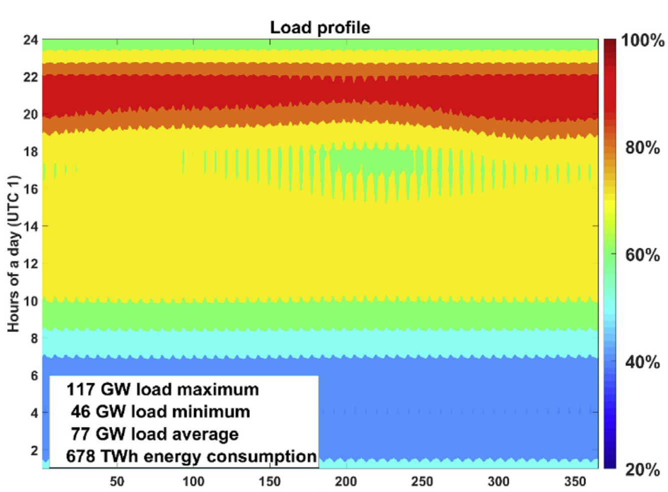

Fig. 6. Aggregated load curve for West Africa for 2050.

Fig. 9 presents the sub-regional capacities projection for 2050. Solar PV capacities are observed in all sub-regions due to very good solar irradiation across WA as shown in Fig. 9 (top) for the BPS-A. Wind capacities are predominant in Niger and Mali. Hybrid PV-battery systems dominate the power system by 2050. Batteries emerge as the major supporting storage technology for PV. Fig. 9 (bottom) shows the capacity outlook in the CPS-A, which is dominated by gas turbines and hydropower. Nigeria with 53 GW (42%) and 93 GW (29%) dominates the total installed capacity in the CPS-A and BPS-A, respectively. Gas-fired power plants (32%) and hydropower (53%) together make up 85% of the WA planned capacities in the pipeline [30]. Fossil gas is expected to be supplied via the Western African Gas Pipeline to coastal countries: Benin, Ghana, Togo, with extension to Côte d'Ivoire. Additional graphical results on regional capacity and generation outlook can be found in the Supplementary Material (Figs. S2 and S3).

3.2. System flexibility

This section analyses the system flexibility, a vital precondition for the grid integration of high shares of power feed-in from variable renewable electricity (VRE), particularly in BPSs. Solar PV, wind energy and hydropower potential in WA is influenced by the monsoon, which causes seasonal variability. The sources of operational flexibility examined in this research include storage technologies, transmission grid, flexible generators (gas turbines), but also dispatchable RE (hydro reservoir, bioenergy) and generation curtailment. These flexibility components act on different timescales.

3.2.1. Storage need and utilisation

Electrical energy storage units are observed to be valuable in providing flexibility to the power system, particularly in the BPSs due to high penetration of RE as shown in Fig. 10(a–d), visualising the electricity throughput. The residual load demand covered by storage units is 259 TWh (36%) in BPS-A, 256 TWh (35%) in BPSnoCC-A, 333 TWh (44%) in BPS-C and 266 TWh (37%) in BPSnoCC-C by 2050. Contrarily, storage units output in the CPSs is about 20 TWh (3%). Battery storage emerges as the key element of the storage units in terms of output, particularly in the BPSs throughout the transition due to daily charge and discharge. Thermal energy storage (TES) and gas storage outputs appear to be relevant in the BPSs across the countries from 2035 to 2045, respectively. Whereas in the CPSs, TES dominates storage output until 2045, due to CSP installed capacities. Absolute storage throughput numbers under various scenarios examined are provided in the Supplementary Material (Tables S19–S23). Table 4 presents the storage capacity by technology for all scenarios in 2050. Fig. 11 shows the required storage capacities under various scenarios. In terms of storage capacities requirement, gas storage dominates in the BPSs throughout the transition, followed by TES in the CPSs until 2045. The installed storage capacities are lower in the BPSs across the area in comparison to the BPSs across the countries as shown in Fig. 11. While the lower installed capacities of storage technologies in the CPSs due to increasing share of dispatchable thermal generators and hydropower.

3.2.2. Cross-border electricity trade flow

Asides from the role of energy storage units in balancing the temporal mismatch between generation and demand, the transmission network provides further flexibility to the power system. It further helps in balancing large regional differences of generation and demand, and offers strong spatial interconnections. The overall grid utilisation profile depicts the electricity generation and demand pattern under various scenarios, as shown in Fig. 12. Grid

Table 2
Description of the scenarios examined for the ECOWAS power sector transition.

Name	Description
Best Policy Scenario – Country wide (BPS–C)	Carbon emissions costs are applied to forbid new investment in fossil-fueled plants and 100% RE is achieved by 2050. In this scenario, no interconnection exist among the 15 member states. Within countries, applied for Nigeria, interconnections are free for optimisation.
Best Policy Scenario – Country wide (BPSnoCC-C)	This scenario is similar to the above. However, no GHG emissions costs are assumed.
Best Policy Scenario – Area wide (BPS–A)	This scenario is similar to the BPS-C. However, interconnection among member states is assumed with GHG emissions costs.
Best Policy Scenario no GHG cost – Area wide (BPSnoCC–A)	This scenario is similar to the BPS-A. However, no GHG emissions costs are assumed.
Current Policy Scenario – Area wide (CPS–A)	This scenario is designed according to ECOWAS RE targets [3]. The RE capacities were provided until 2030. From 2030 onwards capacity for various technologies are extrapolated until 2050. All sub-regions are interconnected. Coal, gas turbines, internal combustion engines and nuclear power plants supply the rest of the capacities required to meet the demand.
Current Policy Scenario no GHG cost – Area wide (CPSnoCC–A)	The Current Policy Scenario is simulated without GHG emissions costs.

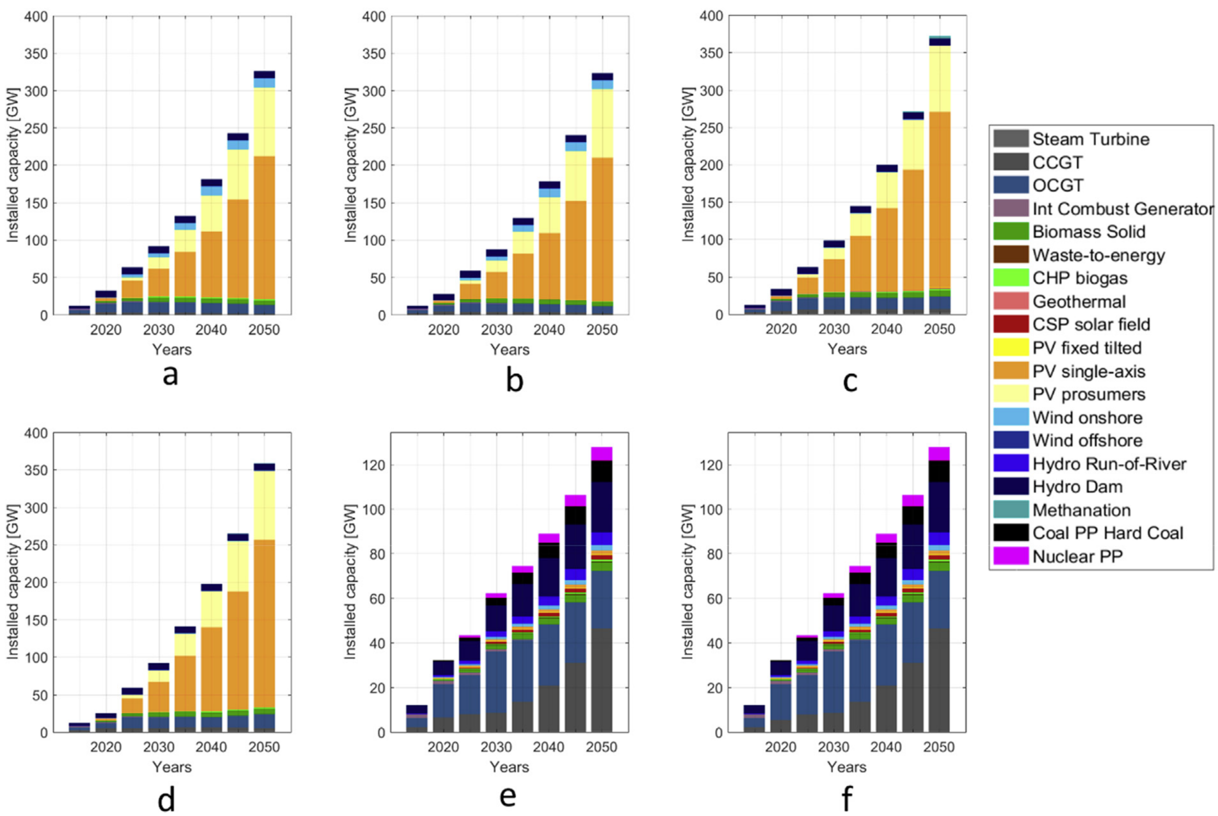

Fig. 7. Installed generation capacities for BPS-A (a), BPSnoCC-A (b), BPS-C (c), BPSnoCC-C (d), CPS-A (e) and CPSnoCC-A (f) from 2015 to 2050.

utilisation appears to be positively related to solar PV generation from around 9.00 to 18.00 per day in the BPS-A, as shown in Fig. 12 (left). Whereas, grid utilisation appears to be more vibrant during the times of higher electricity demand from around 20.00 to 22.00 in the CPS-A, as shown in Fig. 12 (right). As evident in Fig. 13, the 20 sub-regions can be classified as, exporting and importing sub-regions. The annual grid utilisation in the BPS-A is visualised first in Fig. 13 (top). In the BPS-A there are three main exporting sub-regions as shown in Fig. 13 (top); namely, Niger, Mali and NIG-NW. Niger and Mali are net exporters of solar and wind electricity, as a result of high resource potential due to diminishing monsoon influence. While, NIG-NW is a net exporter of solar electricity. Countries with good hydropower potential serve as balancing regions, which include Nigeria (NIG-NE) and Guinea. In the CPS-A, the major exporters are Guinea, Nigeria, Niger and Liberia as shown in Fig. 13 (bottom). The total grid transmission is 298 TWh in the BPS-A, representing 45% of the total electricity demand, and is 39 TWh in the CPS-A, representing 6% of the total electricity demand. The annual grid utilisation increased significantly in the BPS-A by a factor of 7.6 in comparison to CPS-A. Gambia, Sierra Leone and Guinea Bissau did not experience any changes in terms of the amount of electricity imported in the BPS-A and CPS-A. Mali changed from being an importer in the CPS-A to an exporting country in the BPS-A. Electricity imports increased significantly in most of the sub-regions of Nigeria, except NIG-NW, which emerges to be an exporting sub-region in the BPS-A, while NIG-NE maintains status quo in the BPS-A and CPS-A.

Fig. 14 shows the cross-border electricity trade directions and amounts (in TWh) among the sub-regions in the BPS-A and CPS-A. The thickness of the flow illustrates the amount of electricity

Table 3
Electricity installed capacity by technology for all scenarios in 2050.

Generation technology	Unit	BPS-A	BPSnoCC-A	BPS-C	BPSnoCC-C	CPS-A	CPSnoCC-A
PV prosumers	[GW]	92	92	92	92	0	0
PV single-axis tracking	[GW]	192	192	235	223	2	2
PV optimally tilted	[GW]	0	0	1	0	0	0
Wind energy	[GW]	12	12	0	0	2	2
Geothermal power	[GW]	0	0	1	0	0	0
CSP	[GW]	0	0	0	0	2	2
Hydropower	[GW]	10	10	10	10	28	28
Biomass PP	[GW]	5	5	8	7	4	4
Waste PP	[GW]	0	0	0	0	0	0
Biogas PP	[GW]	2	0	2	2	1	1
Biogas Digester	[GW]	1	1	1	1	1	1
Biogas Upgrade	[GW]	1	1	1	1	1	1
CCGT	[GW]	1	2	5	4	46	46
OCGT	[GW]	12	9	17	19	26	26
Steam Turbine	[GW]	0	0	2	1	0	0
Coal PP	[GW]	0	0	0	0	10	10
Oil PP	[GW]	0	0	0	0	0	0
Nuclear PP	[GW]	0	0	0	0	6	6

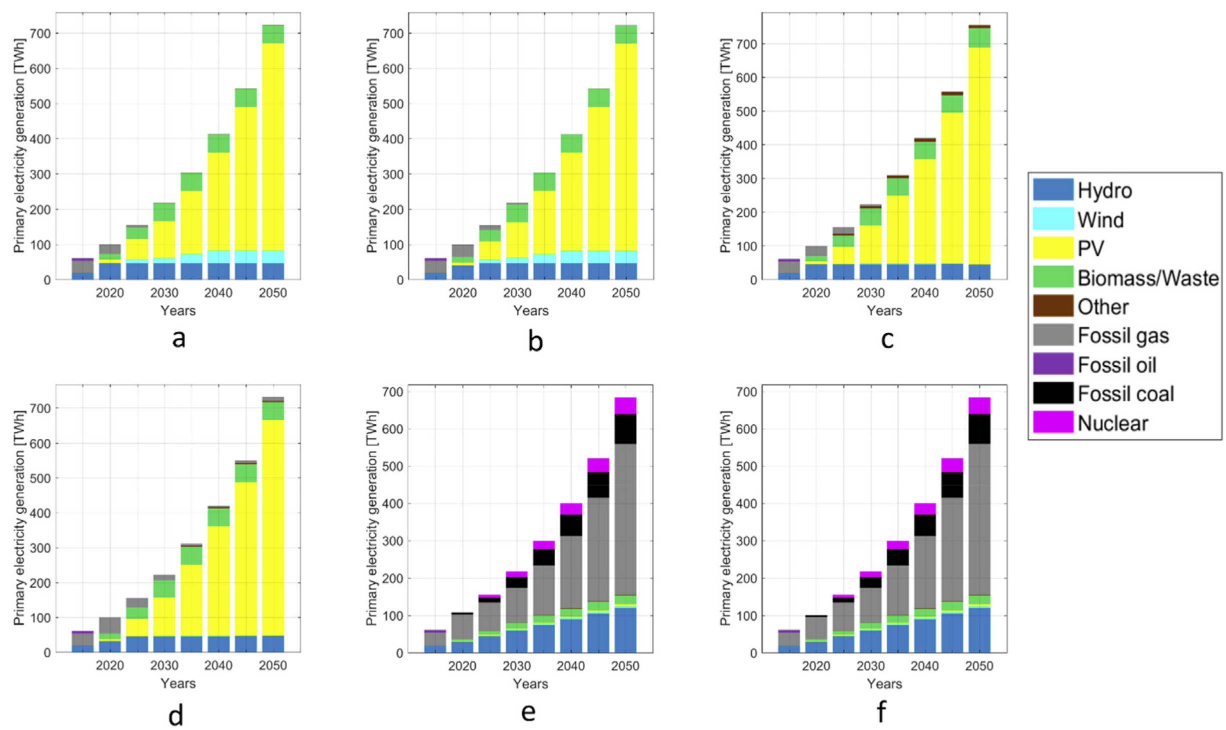

Fig. 8. Electricity generation mix for BPS-A (a), BPSnoCC-A (b), BPS-C (c), BPSnoCC-C (d), CPS-A (e) and CPSnoCC-A (f) from 2015 to 2050.

transferred between the regions in TWh. Niger dominates the net electricity trade with 204 TWh (68%), followed by 54 TWh (18%) for Mali and 34 TWh (11%) for NIG-NW in BPS-A. Whereas, Guinea dominates the electricity trade in the CPS-A with 12 TWh (31%). Benin and NIG-NC emerge as the main transit conduit in the regional electricity trade in the BPS-A, including Gambia and Guinea Bissau in the CPS-A as shown in Fig. 14 (bottom). The regional electricity trade depicts the role of grid interconnections, particularly in scenario with high shares of RE, in comparison to the one dominated by thermal dispatchable generators.

3.2.3. *Role of gas turbines*

The fleet of VRE generators are accompanied by flexible power plants. Gas turbines are found to be dynamic fast-responding dispatchable generators, covering some fraction of the residual load demand based on different timescales from days to weeks, particularly in the BPSs. The average FLH declined from 5470 in 2015 to 191 in 2050 for BPS-A, to 275 in BPSnoCC-A, to 417 in BPS-C and to 601 in BPSnoCC-C. This document a drastic shift in the role of gas turbines, from bulk electricity generation to peak electricity generation for the most challenging seasons of the year. By 2050, gas turbine generation is 2.5 TWh in BPS-A, 3.1 TWh in BPSnoCC-A, 9.0 TWh in BPS-C and 14.0 TWh in BPSnoCC-C. The installed gas turbine capacities are 12.9 GW, 11.4 GW, 21.7 GW and 23.3 GW BPS-A, BPSnoCC-A, BPS-C and BPSnoCC-C, respectively, by 2050. Whereas in the CPSs, the CCGT operates as baseload generators and OCGT is utilised in meeting peak loads during the night. Fig. 15 illustrates the gas turbine (CCGT and OCGT) utilisation for BPS-A, BPS-C and CPS-A by 2050. Gas turbines are most utilised during the monsoon period in BPSs as shown in Fig. 15 (a-b, d-e).

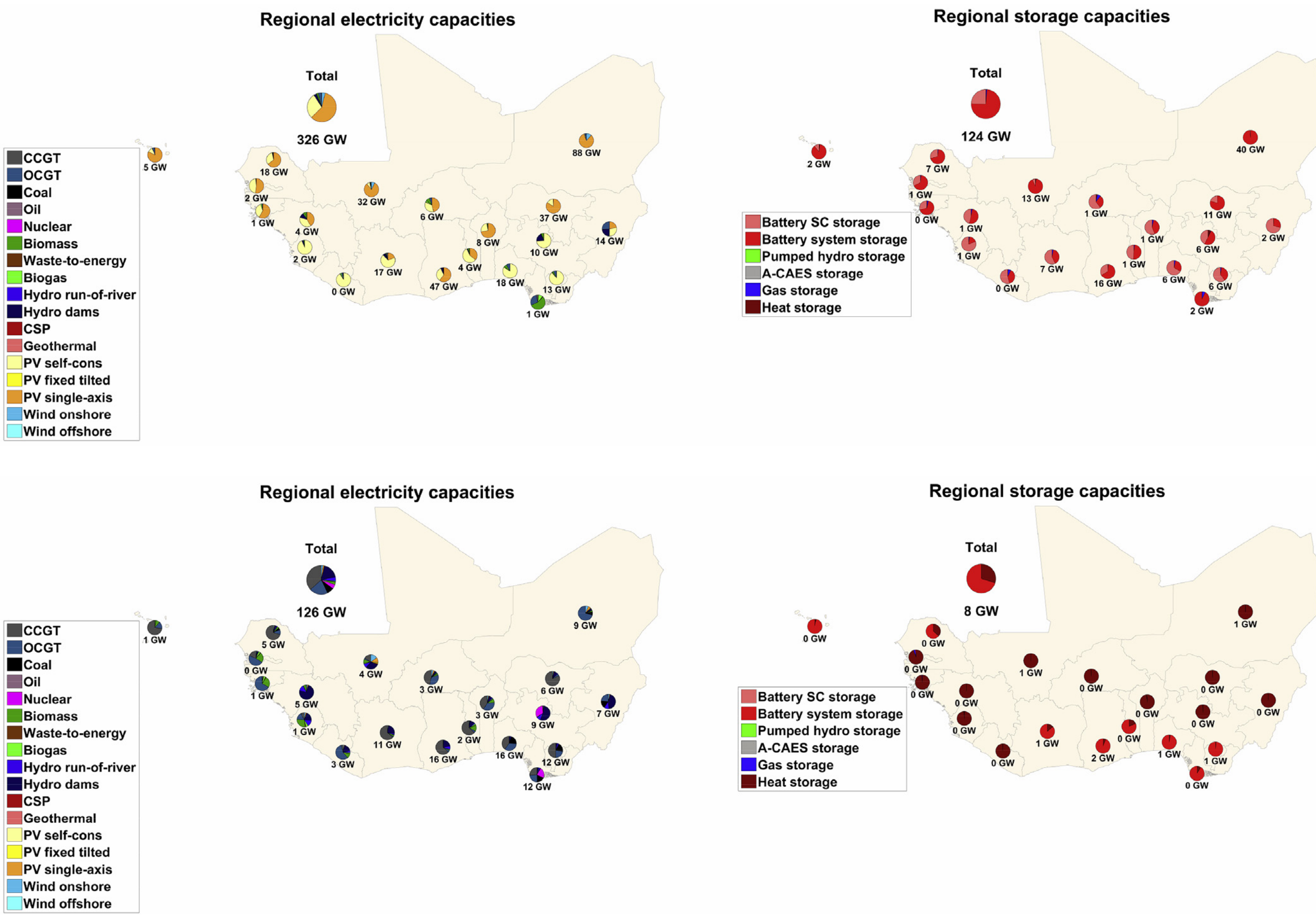

Fig. 9. Overview of the generation mix (left) and energy storage (right) for the BPS-A (top) and CPS-A (bottom).

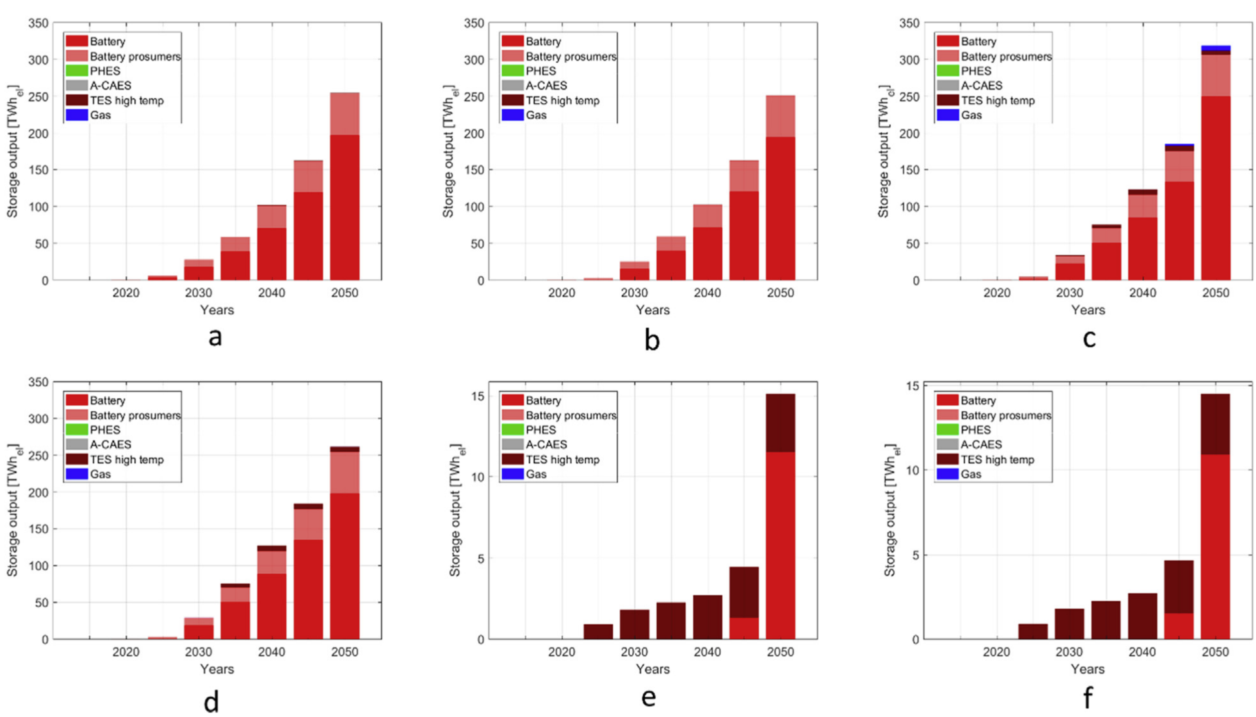

Fig. 10. Storage throughput for BPS-A (a), BPSnoCC-A (b), BPS-C (c), BPSnoCC-C (d), CPS-A (e) and CPSnoCC-A (f) from 2015 to 2050.

Table 4
Storage capacity by technology for all scenarios in 2050.

Storage technology	Unit	BPS-A	BPSnoCC-A	BPS-C	BPSnoCC-C	CPS-A	CPSnoCC-A
Battery total	[GWh]	735	737	840	800	33	31
PHS storage	[GWh]	0	0	0	0	0	0
TES storage	[GWh]	5	3	164	49	19	19
A-CAES storage	[GWh]	0	0	7	11	0	0
Gas storage	[GWh]	2403	1738	9359	1630	8	9

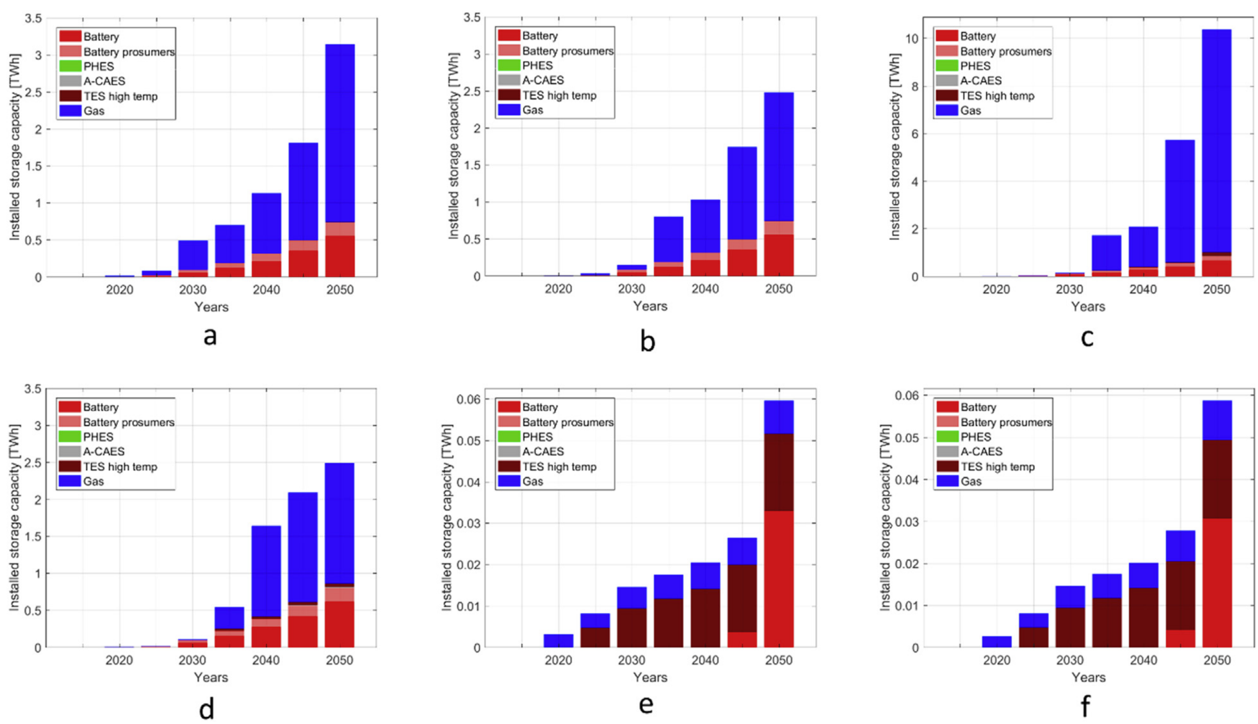

Fig. 11. Installed storage capacities for BPS-A (a), BPSnoCC-A (b), BPS-C (c), BPSnoCC-C (d), CPS-A (e) and CPSnoCC-A (f) from 2015 to 2050.

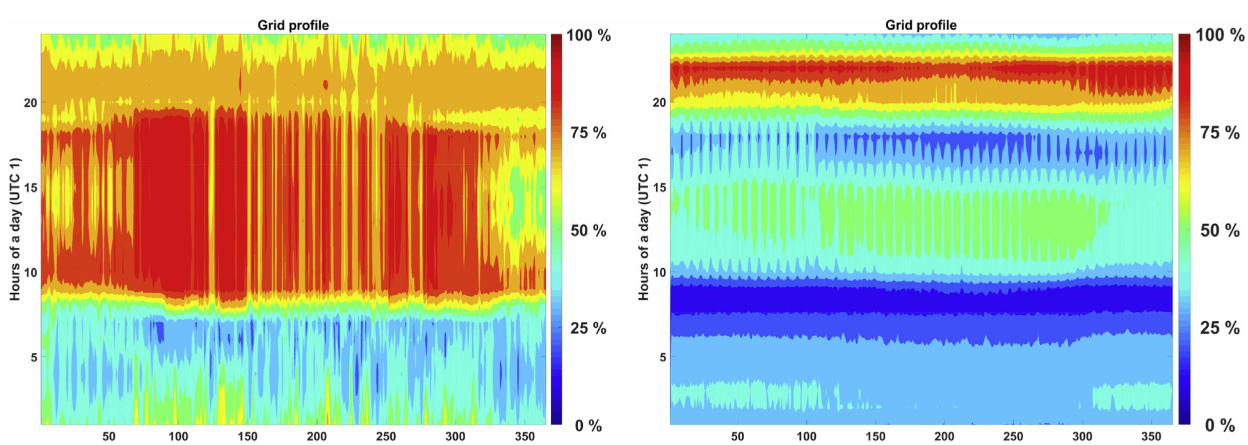

Fig. 12. Grid utilisation profiles for BPS-A (left) and CPS-A (right) in 2050.

Additionally, gas turbines are flexible power plants because their power output can be modulated sufficiently fast when needed and the robustness of their market power setup.

3.2.4. Generation curtailment

Curtailment is a flexibility option, which can be low cost in highly renewable energy systems, compared to other flexibility options [49]. In addition, sudden and unexpected power imbalance in the system is controlled by partial generation curtailment. VRE generation and curtailment under various scenarios are shown in Fig. 16. Fig. 16 (right) further shows the curtailed generation potential corresponding to VRE generation. The curtailed generation potential increases throughout the transition in BPSs, with the exception of the CPSs. It can be observed that generation

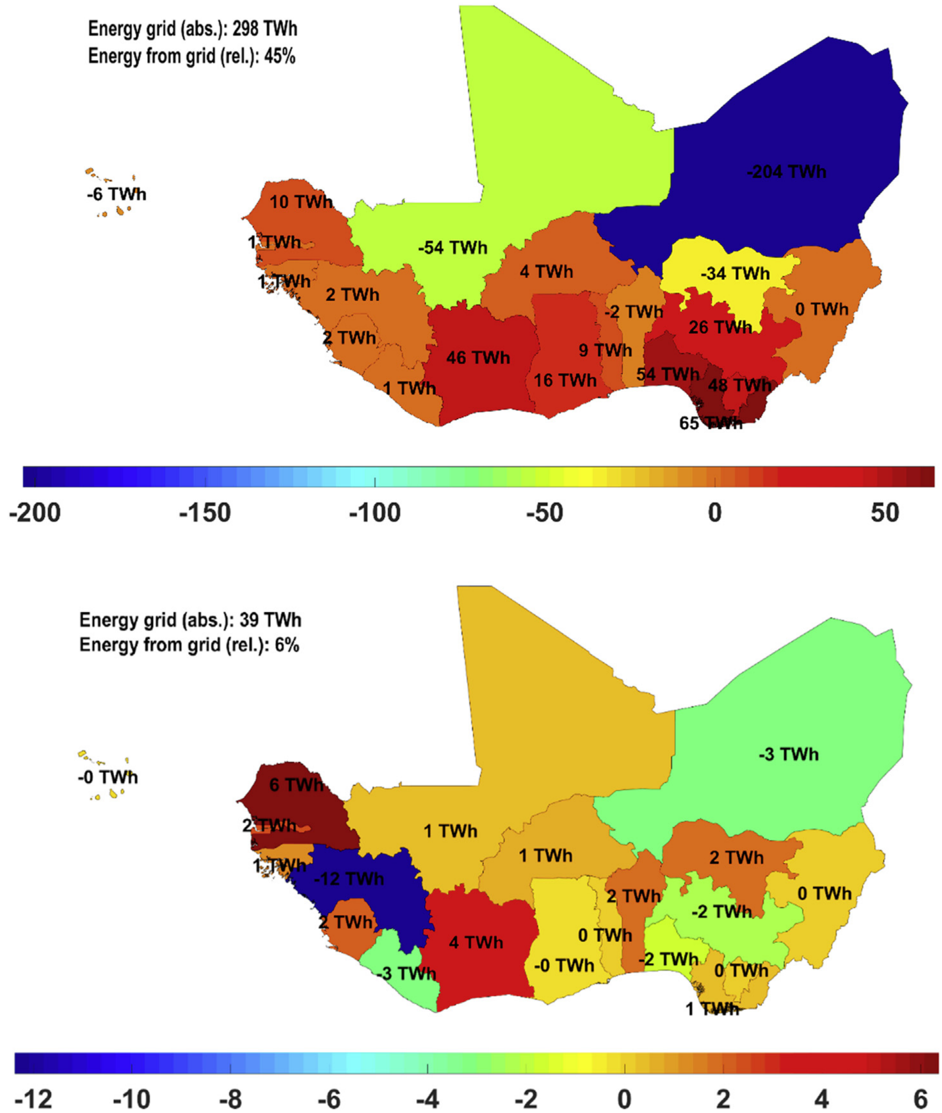

Fig. 13. Absolute annual grid utilisation in the BPS-A (top) and CPS-A (bottom) for the ECOWAS region in 2050. Net annual exports is represented by negative values, while the positive values represent the net annual imports.

curtailment is higher in the BPSs across the countries than in BPSs across the area. The high diversity of the region's power mix balances the regional electricity generation and contributes to low curtailment in the BPSs across the area. By 2050, generation curtailment is 27 TWh (4.1%) in BPS–A, 28 TWh (4.1%) in BPSnoCC–A, 41 TWh (6.1%) in BPS–C, 39 TWh (5.8%) in BPSnoCC–C and 3 TWh (0.5%) in CPSs.

3.3. Energy flow overview

Fig. 17 illustrates the energy flows for 2015 and the BPS in 2050. It shows the generation, end use and efficiency of each energy conversion process. The produced heat and losses in the system consist of the difference between the primary power generation and final demand. In 2015, electric power generation is provided mainly by gas turbines and hydropower, while RE generation, particularly solar PV, dominate in 2050.

3.4. Financial implications of the energy transition

Fig. 18 illustrates the LCOE under various scenarios during the transition. The LCOE in the BPSs is visualised first as shown in Fig. 18(a–d). The LCOE decline from about 70 €/MWh in 2015 to 36 €/MWh in BPS-A, to 41 €/MWh in BPS-C by 2050. Whereas, LCOE declines from 68 €/MWh in 2015 to 36 €/MWh in BPSnoCC-A and to 39 €/MWh in BPSnoCC-C by 2050. Lower LCOE over time during the transition in BPSs signifies the cost competitiveness of RE technologies during the transition. Fig. 18(e–f) depicts the LCOE in the CPSs. The LCOE increased significantly from about 70 €/MWh in 2015 and to 116 €/MWh in CPS-C by 2050, while the LCOE increased slightly from 68 €/MWh in 2015 to 70 €/MWh in CPSnoCC by 2050. Fuel costs and GHG emissions costs contribute to the high LCOE in the CPSs. Additional results on costs are provided in the Supplementary Material (Table S18 and Figs. S4–S6).

3.5. Socio-economic prospects of the energy transition

3.5.1. Jobs in the West African power sector

Figs. 19 and 20 depict the direct energy jobs created in the WA power sector during the transition period for the BPS and CPS. The annualised jobs created in the BPS and CPS were estimated based on the methodology presented in Ref. [33,34] and the assumed

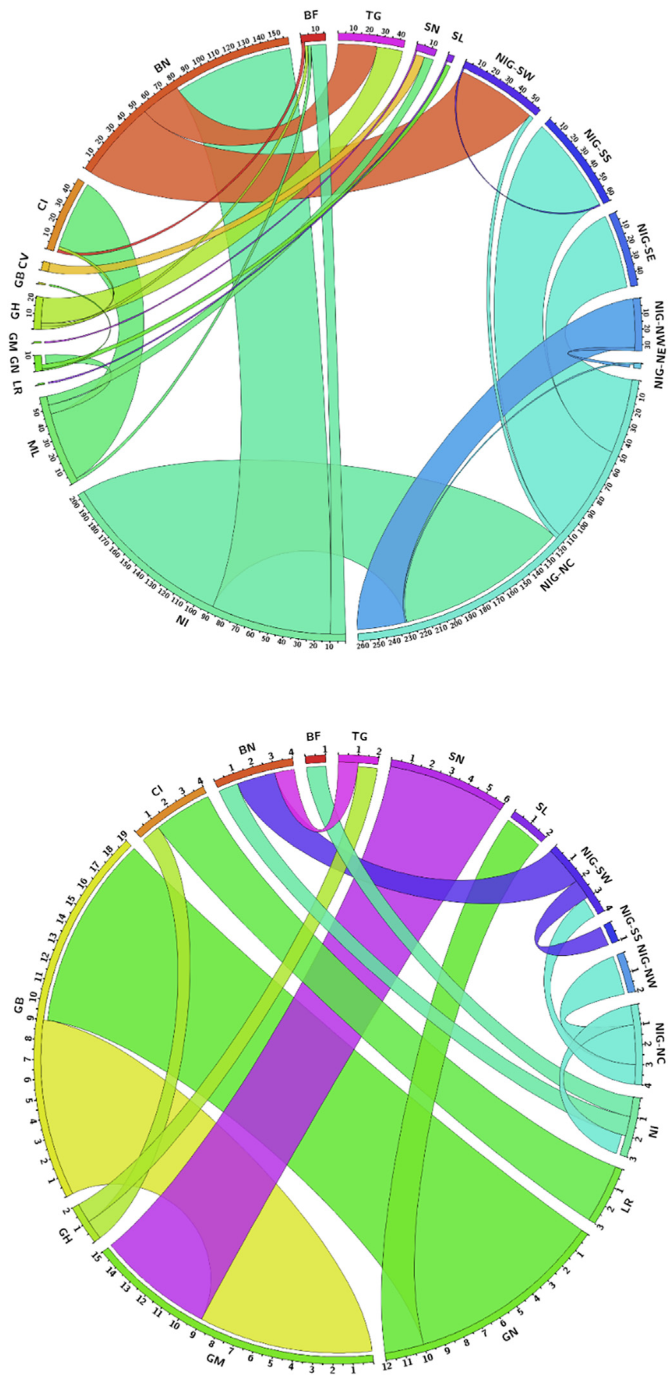

Fig. 14. Regional grid exchange within West Africa in the BPS-A (top) and CPS-A (bottom) in 2050.

employment generation factors are presented in the Supplementary Material (Table 24). About 440 thousand direct energy jobs were created in the BPS and solar PV emerges as the prime job creator with 68% of the total jobs created by 2050, as shown in Fig. 19. Whereas, fossil gas based power generation creates more jobs in the CPS, with 45% of the jobs by 2050, as shown in Fig. 20. Batteries mainly drive jobs created by storage technologies from 2025 onwards, which increases to 44 thousand by 2030 and further increases to 76 thousand by 2050 in the BPS. Overall, number of direct energy jobs created in the BPS appears to grow massively from around 20 thousand in 2015 to about 440 thousand in 2050. Similarly, jobs created in the CPS increase from around 20 thousand in 2015 to about 183 thousand by 2050.

Furthermore, the distribution of jobs by categories during the transition in the BPS and CPS is indicated in Figs. 19 and 20. In the BPS, construction and installation of renewable energy technologies create the bulk of the jobs, enabling the rapid ramp-up of capacity until 2025. The sector creates about 82 thousand jobs by 2050. Furthermore, manufacturing jobs have a relatively low share until 2020, due to higher share of imports (mainly PV, hydropower, bioenergy and fossil gas). From 2025 onwards, a higher share of local manufacturing jobs is observed, as domestic production capabilities are assumed to be increased until 2050. The operation and maintenance jobs appear to take the lead from 2035 onwards and become the main job sector by 2050 with 41% of the total jobs. On the contrary, in the CPS, fuel jobs dominate during the transition. Fuel jobs dominate with a share of 56%, followed by operation and maintenance with 25% while manufacturing, construction and installation jobs are 16% by 2050.

The least-cost option becomes the cornerstone for a vibrant and job-intensive policy option for WA, centred on decarbonisation of its power sector. The electricity demand specific jobs in the BPS increases significantly from 321 jobs/TWh_{el} in 2015 to 1238 jobs/TWh_{el} in 2025 with the rapid ramp-up in RE installations. Beyond 2015, it declines steadily to around 662 jobs/TWh_{el} in 2050 as shown in Fig. 19. Whereas, the electricity demand specifics jobs in the CPS increased from 321 jobs/TWh_{el} in 2015 to 506 jobs/TWh_{el} in 2020, and further declines to 274 jobs/TWh_{el} in 2050.

3.5.2. *Greenhouse gas emissions trajectory under various transition scenarios*

The GHG emissions trajectory under various scenarios are depicted in Fig. 21. The defossilisation of the WA power system appears to be rapid in the BPSs area in comparison to the BPSs country as illustrated in Fig. 21(a–d). With GHG emissions costs, emissions decline rapidly after 2025 as fossil natural gas plants are replaced with RE capacities as observed in BPS-A and BPS-C. Without GHG emissions costs, the BPSnoCC-A shows similar emission pattern with BPS-A, whereas the BPSnoCC-C deviates the emission pattern observed in BPS-C. In the CPSs, the GHG emissions increased throughout the transition as depicted in Fig. 21(e–f).

4. Discussion

This work investigates the least-cost electricity expansion option for WA, under certain policy constraints. The BPSs results show that deep-decarbonised pathways are cheaper than fossil-based CPSs. Such a system complies with the objectives set out in the Paris Agreement, in comparison to CPSs. This power system is achievable with the abundant and diverse RE resources in WA, but requires a strong political will.

4.1. *Solar PV plus battery: the new workhorses of the West African power system*

Solar PV emerges as the new workhorse of the WA power system in renewables-led electricity generation. A mix of RE technologies, as observed in the BPSs can substitute the current heavy dependence on gas-fired electricity. The power system optimisation results show that it is least-costing to supply 81%–85% of the electricity demand in WA from solar PV under the BPSs by 2050. In this cost-optimal expansion path, about 284–328 GW solar PV installed capacities are operational by 2050. Solar PV technologies supply 589–644 TWh in the BPSs. PV prosumers contribute 20% to the total electricity generation in 2050. The plausible reasons for a high share of PV in WA power system includes the following, the

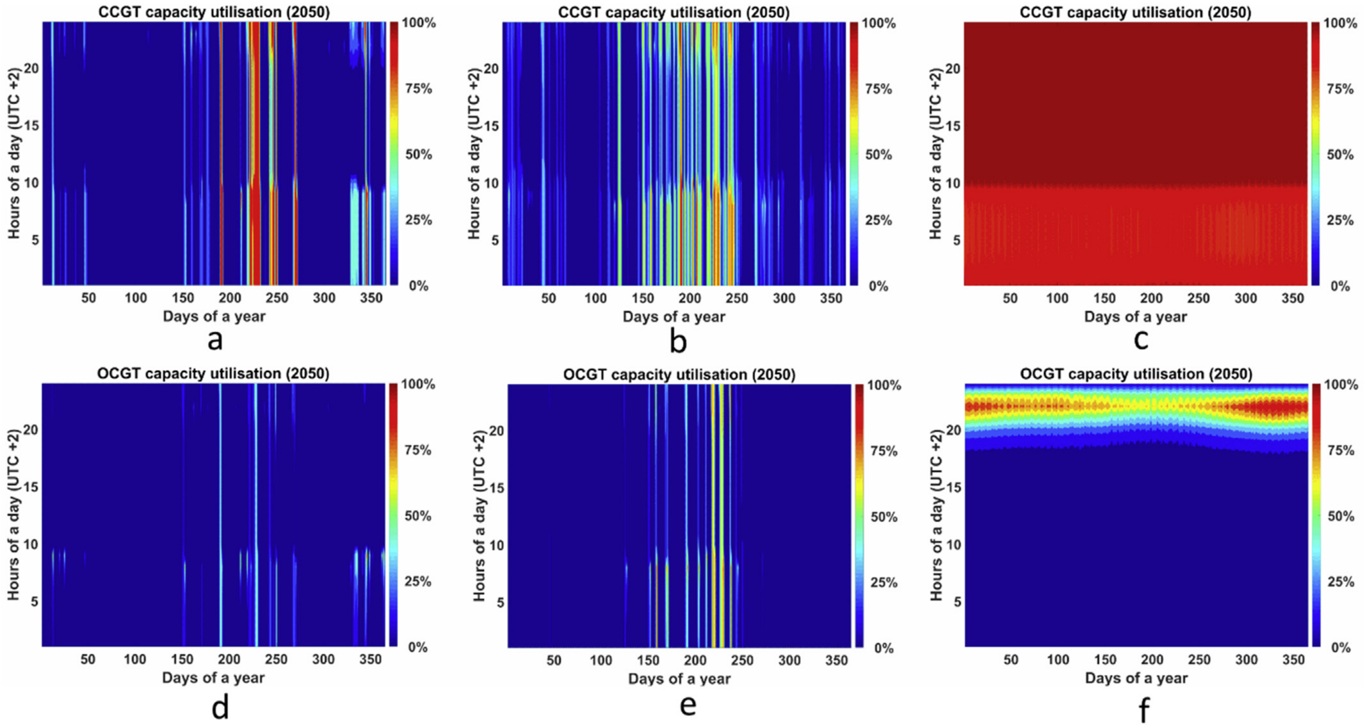

Fig. 15. Combined cycle gas turbine capacity utilisation for BPS-A (a), BPS-C (b), CPS-A (c). Open cycle gas turbine capacity utilisation BPS-A (d), BPS-C (e), and CPS-C (f) in 2050.

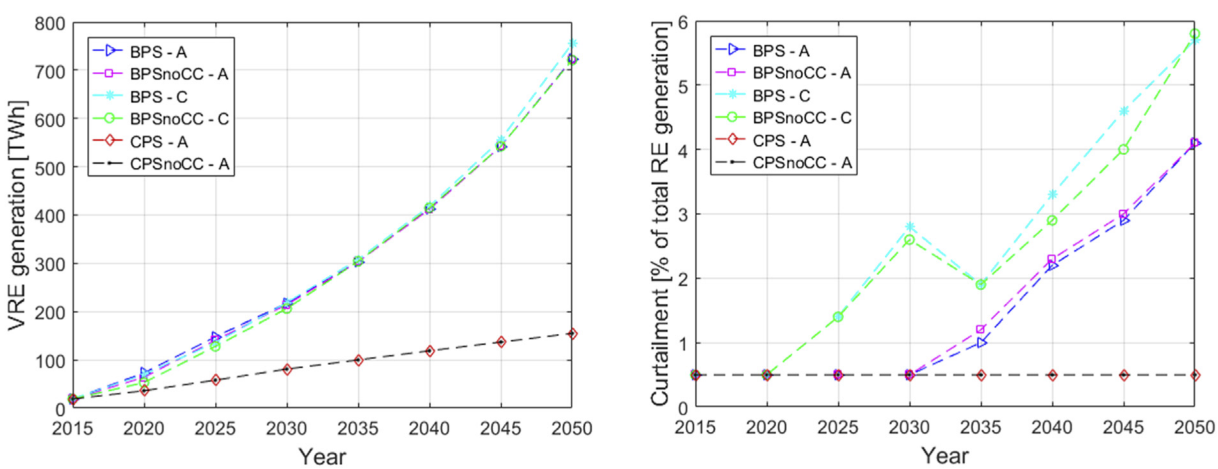

Fig. 16. Variable RE generation (left) and curtailment of generation potential (right) in TWh under various scenarios during the transition.

steep fall in solar PV and battery costs [37], excellent resource conditions and fast growing electricity demand in WA. The PV-battery combination drives most of the system from 2030 onwards. Batteries emerge as the main supporting technology for PV, battery capacities ranged from 735 GWh to 840 GWh in the BPSs. The solar resource potential increases northward due to diminishing monsoon influence. Likewise, highest wind potential is found towards the north/north-west. These factors contribute to high solar PV and wind energy installed capacities in Niger and Mali. WA has the solar and land resources to technically host a power system led by a mix of RE technologies. However, assessments of solar PV and wind power mixes for WA are rare [4]. The required land area for solar PV and wind installation is estimated based on the specified capacity density assumed in the model, which is 8.4 MW/km^2 and 75 MW/km^2 for onshore wind and optimally tilted PV respectively [32]. Therefore, an area of 1454 km^2 and 3784 and is needed for wind and solar PV capacities by 2050, representing just 0.03% and 0.07% of the total land area of WA. The results of this research reveal that solar PV will emerge as the backbone of WA power system, which is similar to the results of Oyewo et al. [19] for Nigeria. They conclude that solar PV will contribute significantly to Nigeria's future power system and it is comparable to any other developing countries with similar climates. Adeoye and Spataru [31] show that the high integration of solar PV plants will reduce the supply-demand gap and load shedding in the WA by 2025. According to Ref. [31], 71 GW of solar PV plants is assumed to be installed and in operation across WA. The BPSs results are also similar to results of Barasa et al. [26] for entire Sub-Saharan Africa (SSA) based on an overnight scenario approach for 2030 conditions, they conclude that countries in SSA

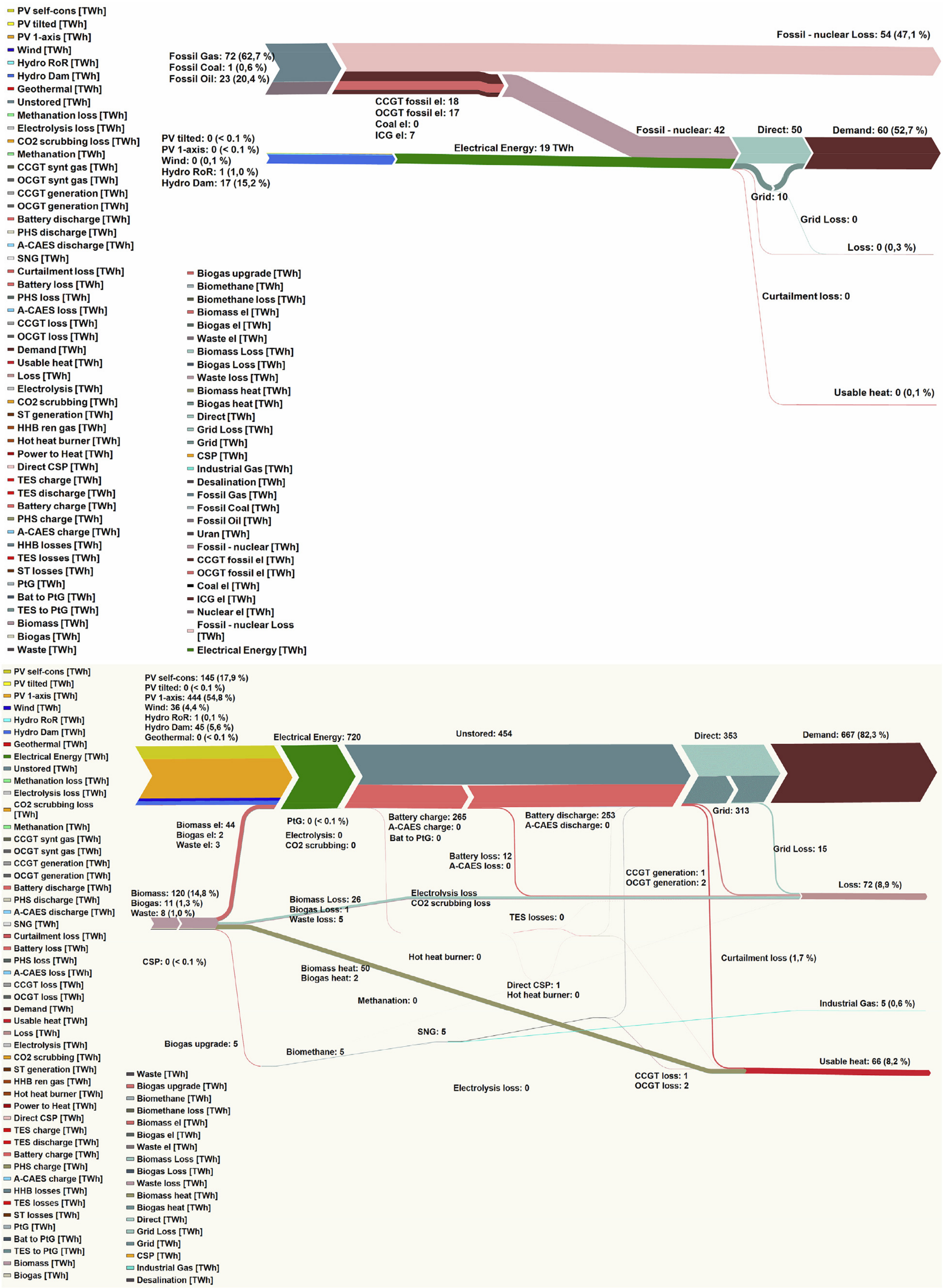

Fig. 17. Energy flows of the system for 2015 (top) and BPS in 2050 (bottom).

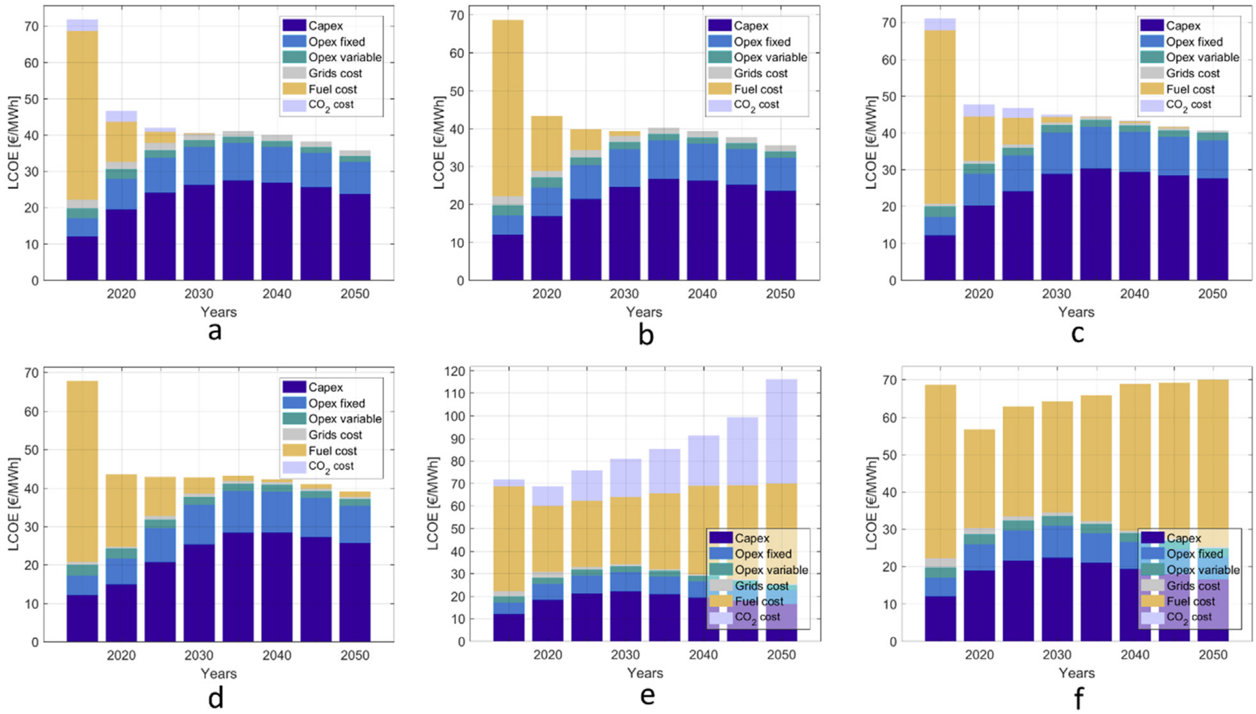

Fig. 18. Levelised cost of electricity for BPS-A (a), BPSnoCC-A (b), BPS-C (c), BPSnoCC-C (d), CPS-A (e) and CPSnoCC-A (f) from 2015 to 2050.

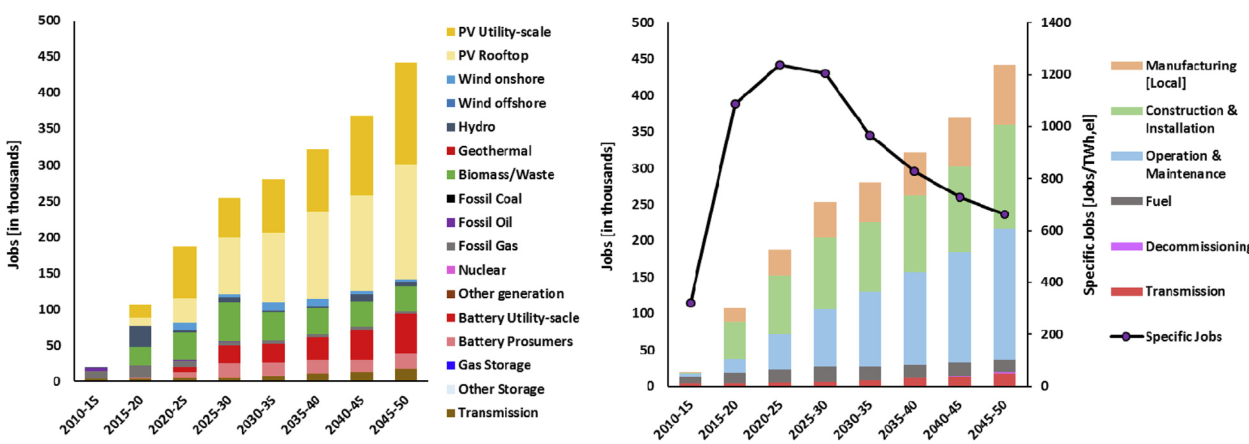

Fig. 19. Jobs created by the various power generation and storage technologies (left) and jobs created based on different categories with the development of electricity demand specific jobs (right) during the energy transition from 2015 to 2050 in West Africa for the BPS.

can be powered majorly by wind energy and solar PV. A recent study by IRENA [30], shows that the grid connected solar PV market would be over 20 GW by 2030, compared to just over 8 GW in their Reference Scenario (RS). The IRENA National targets scenario achieved a share of 37% RE. According to Ref. [30], solar PV installed capacity is 21.5 GW, wind is 1.6 GW and hydropower is 11.5 GW by 2030, the remaining generation capacity supplied by fossil generators is dominated by gas turbines with about 30 GW. In the CPSs, fossil-based thermal generators and hydropower dominate the power system accounting for 532 TWh (78%) and 120 TWh (18%) respectively by 2050. The results of the CPSs are comparable to the New Policy Scenario (NPS) of the IEA [17]. According to Ref. [17], fossil-based thermal generators dominate with 323 TWh (68%) of which, gas supply is 249 TWh (53%), followed by hydropower with 100 TWh (21%), bioenergy with 12 TWh (3%), solar PV with 17 TWh (4%) and other renewables with 21 TWh (4%). Similarly, gas and hydropower are expected to roughly triple in the IRENA Reference Scenario by 2030 to 32 GW and 11.5 GW, respectively [30]. It is worth mentioning that despite the huge potential for CSP as alternative technology to harness the Sun's energy in WA, future costs appear to be a bottleneck for CSP development. CSP installed capacities ranged from 0.1 to 1 GW, the highest installed capacities occur in the CPSs and lowest in the BPS across the countries, whereas no CSP is installed in the BPS across the area. Overall, the ECOWAS Renewable Energy Policy (EREP) targets for grid-connected capacity for solar PV, wind energy and CSP by 2030, in WA is very low at 1 GW for each [3]. It is obvious that solar PV and wind energy will be very relevant in a least-cost expansion for WA future power sector, while policies to facilitate their deployment are exigent.

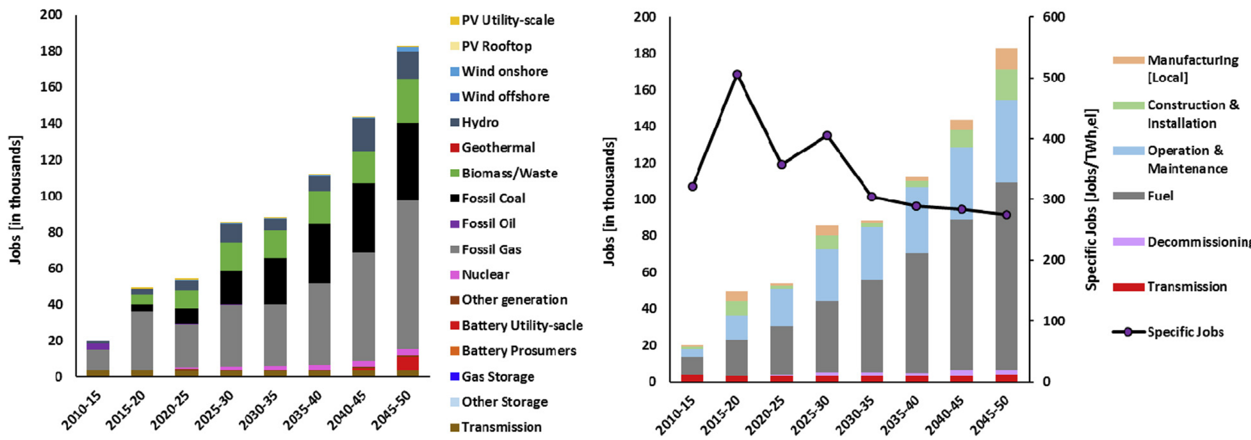

Fig. 20. Jobs created by the various power generation and storage technologies (left) and jobs created based on different categories with the development of electricity demand specific jobs (right) during the energy transition from 2015 to 2050 in West Africa for the CPS.

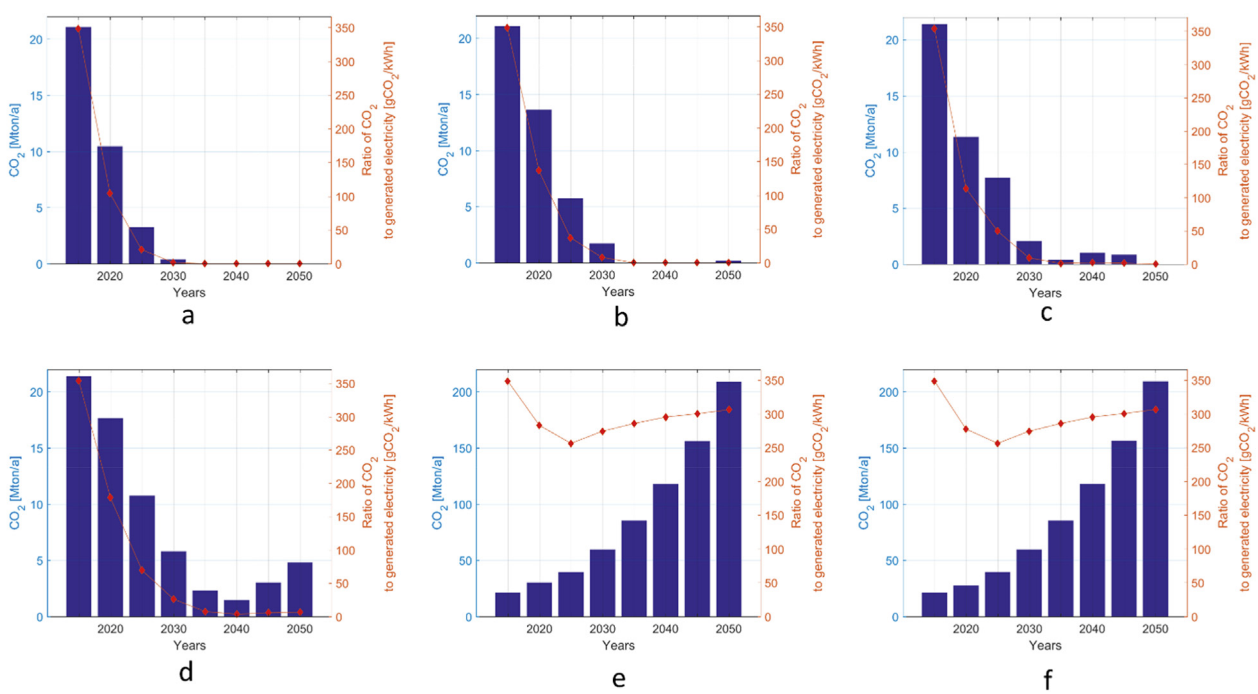

Fig. 21. GHG emissions trajectories for BPS-A (a), BPSnoCC-A (b), BPS-C (c), BPSnoCC-C (d), CPS-A (e) and CPSnoCC-A (f) from 2015 to 2050.

4.2. Flexibility requirement

Flexibility is harnessed from dispatchable renewables (mainly biomass and hydropower), transmission grid, curtailment and storage technologies, which facilitates the high shares of solar PV as seen in the BPSs. The relevance of storage technologies appears to be more vibrant in the BPSs, due to increasing shares of solar PV in the national power systems. Power grids dominated by solar PV are typically characterised by high storage needs [27]. In the BPSs, the relevance of storage technologies, particularly battery storage, increases over time. The PV-battery hybrid system appears to be a least-cost option for the WA power system by 2050, which is similar to the results of Oyewo et al. [27] for the Nigerian power system. Similar results have been obtained for the competitiveness of hybrid PV-battery plants for the case of Morocco [50]. In addition, battery cost has declined over the years, and further cost reduction is expected [51,52]. In terms of storage capacities requirement, gas storage leads in all the BPSs followed by battery storage. However, battery storage dominates when storage throughputs are considered, due to daily requirement in the absence of solar PV. Several studies have shown a need for storage, once the share of RE power generation reaches approximately 50% [28,32,37,49]. Consequently, with a share of 80% RE, seasonal storage is required [32]. The long term and seasonal storage is compensated by TES, adiabatic compressed air energy storage (A-CAES) and power-to-gas (PtG). Seasonal storage contribution is not required in the BPS across the area in comparison to BPS across the countries. Rather, bioenergy and hydropower are sufficient in compensating the seasonal variation in the BPS across the area, along with the vital flexibility provided by cross-border interconnections. As a result, only 9.5 GW PtG conversion units are needed in the BPS across the countries, whereas no PtG conversion units are needed in the BPS across the area. This rather rare phenomenon is also observed for Brazil, based on an overnight scenario approach for 2030 [53]. According to Ref. [53], a 100% RE system can be operated with extremely low seasonal storage based on PtG, this special condition is traceable to very high shares of dispatchable hydropower reservoirs and equatorial weather conditions. Hydropower reservoirs could be considered as virtual batteries, if operated in conjunction to solar PV systems [54]. The extraordinary growth of wind energy, plus some PV and bioenergy in Uruguay can be attributed to its massive hydropower, which can cover shortfall in wind, plus cross-border interconnections [55]. Under the right circumstances, as seen in the BPS area, a mix of RE appears to be a good candidate for the WA power sector. The synergies of RE resources limit the need for storage, improve system flexibility, and reduce day-to-day and inter-annual variability of the system as observed in the BPS area.

Furthermore, the flexibility function of the grid appears to be more vibrant in the BPS across the area for counterbalancing the regional power mismatches. Graphical results on grid utilisation are presented in the Supplementary Material (S7). Grids are a very important flexibility option, and their function will be discussed in more detail in the next section. Curtailment is another flexibility option. Curtailment is widely expected to increase with more VRE [49,56,57], as observed in this research. The question, whether to avoid curtailment or not, is debated increasingly [56], based on the perception that curtailment would be a mere waste of energy, despite the reported technical benefits [49]. Conversely, curtailment is another valuable resource, helping to stabilise the grid and improve system flexibility [57]. Avoiding curtailment requires investing in grid interconnection and storage solutions [56]. Contingencies in the power system can be reduced by curtailment of generation, which appears to be higher in the BPS across the countries than in the BPS across the area. The curtailment-storage-penetration nexus in the energy transition is researched in Ref. [49]. A set of two scenarios were examined in Ref. [49], the first scenario anticipates a limit on curtailment, while the other expects an optimal transition with curtailment considered as a technical solution to minimise mismatch in demand and supply. According to Ref. [49], "curtailment in optimally managed energy transition brings techno-economic opportunities to the system, while no curtailment or improper use of curtailment carriers a penalty".

The WA electricity generation depicts a tendency towards flexibility and complementarity as observed particularly in the BPSs. The need for flexibility will continue to increase in systems dominated by RE such as BPSs, as solar PV appears as the default technology choice for low-cost bulk energy in the power sector, in particular in Sun Belt countries [26,28,37,49,50]. Based on the foregoing discussion, gas power plants appear to be a valuable and flexible peaking technology when required, particularly during the monsoon periods. Several studies have identified gas turbines as a flexibility option for systems dominated by high RE shares, instead of coal or nuclear [19,28,57,58]. A recent study concludes that required flexibility could be provided by using flexible natural gas fired turbines, if the costs of battery do not decrease [59]. Furthermore, there are strong signals that electricity will contribute significantly to the future energy system. An electricity-led energy system will reduce primary energy demand and further increase system flexibility by coupling the decarbonised electricity to heat, transport and industrial sectors [25,60]. What is more, PtG and power-to-liquids solutions can be used to supply residual demand for fuels and chemicals that cannot be replaced with electricity directly [25,60].

4.3. Benefits of cross-border electricity trade

The interconnection of sub-regions in WA offers multiple benefits. Broad cost saving opportunities and increased flexibility in the power system is achieved due to grid interconnections between sub-regions of WA. The transmission network facilitates the exploitation of best RE production sites and smoothens out day-to-day variability. The transmission interconnections have the potential to reduce the LCOE, total system cost, installed capacities, storage requirements and curtailment in a highly renewable WA power system, as observed in the BPS across the area. With GHG emissions costs, the LCOE drops to 40.6 €/MWh in the BPS across the countries and to 35.7 €/MWh in the BPS across the area by 2050. Without GHG emissions costs, LCOE decreases to 39.0 €/MWh in the BPS across the countries and to 35.5 €/MWh in the BPS across the area. Likewise, the total annualised cost of system is 27.2 b€ in the BPS across the countries and it is 23.9 b€ in the BPS across the area with GHG emissions costs. Without GHG emissions costs, the total annualised cost of system is 26.1 b€ in the BPS across the countries and it is 23.8 b€ in the BPS across the area. This implies annual cost savings of 9% or 2.3 b€/a in the BPS across the area without GHG emissions costs, and is 12% or 3.3 b€/a with GHG emissions costs. This result requires special highlighting: the BPS across the area with and without GHG emissions costs lead practically identical energy system solutions. Table 5 highlights the key differences in financial outcomes and selected electricity parameters for 2050. It can be seen that the system costs are more or less the same as shown in Table 5 for the BPSs, which implies that with or without GHG emissions costs, the WA power sector can reach 100% RE as the least cost solution. What is more, the BPSs without GHG emissions costs leads to 98% and 99% RE generation in the area and country scenarios, respectively.

With transmission and GHG emissions costs, the BPS across the area shows approximately 12% less installed capacities (327.9 GW vs 370.8 GW), 70% less storage capacity (3.1 TWh vs 10.4 TWh) and 22% less storage output (259.2 TWh vs 332.9 TWh) than the BPS across the countries. Likewise, with transmission and no GHG emissions costs, the BPS across the area shows a reduction of about 10% of installed capacities (325.2 GW vs 360.5 GW), 11% less energy storage capacity (2.4 TWh vs 2.7 TWh) and 4% less storage output (256.2 TWh vs 266.5 TWh) compared to the BPS across the countries. Faster defossilisation of the power system can be achieved as observed in the BPS across the area in comparison to the BPS across the countries. The remaining cumulative GHG emissions are approximately 37% lower (75 $MtCO_{2eq}$ vs 119 $MtCO_{2eq}$) in the BPS across the area with GHG emissions costs, without GHG emissions costs it is about 51% lower (109 $MtCO_{2eq}$ vs 221 $MtCO_{2eq}$) both compared to the BPS across the countries and aggregated from 2018 to 2050.

Investments in the transmission and distribution (T&D) infrastructure are vital for any emerging power grid system, such as WA. For instance, to facilitate the export of wind production from Niger and Mali as observed in the BPS across the area from 2025 onwards, the development of the North-core interconnection linking Niger, Nigeria, Benin and Burkina Faso is significant. According to IRENA [30], the development of nearly all cross-border transmission projects between WA countries currently in the pipeline appear to be beneficial. The World Bank (WB) estimates the economic benefits of the WAPP at 5–8 bUSD per year. Through the International Development Association (IDA), 750 mUSD of current WB support is directed towards the completion of the primary interconnections in WA. About 4000 km of grid lines are currently under development, all to be completed in the early 2020s [13]. According to the IEA NPS [17], the cumulative investment in T&D lines for the WA power sector is 229.3 bUSD from 2014 until 2040.

4.4. Off-grid electrification

The development of T&D infrastructure is recognised as a solution to the challenges of low access and unstable power

Table 5
Differences in key energy system parameters and financial results in 2050 under various scenarios.

	unit	BPS-A	BPSnoCC-A	BPS-C	BPSnoCC-C	CPS-A	CPSnoCC-A
Total annual levelised cost	[b€]	23.9	23.8	27.2	26.1	78.9	47.5
Levelised cost of electricity	[€/MWh,el]	35.7	35.5	40.6	39.0	116.2	70.0
Generation	[TWh,el]	723.2	722.7	754.7	732.5	683.4	
Installed capacity	[GW]	327.9	325.2	370.8	360.5	129.7	
Storage capacity	[TWh,el]	3.1	2.5	10.4	2.7	0.06	0.06
Curtailment	[%]	4.1	4.1	6.1	5.8	0.5	

supply in the region [17]. According to Ref. [17], most of the investments in electrification expansion in SSA go towards on-grid access, new T&D lines account for more than half of the investments. WA accounts for the largest share of the investments with around 75 bUSD (37%). Bertheau et al. [61] investigate the impacts of grid expansion in SSA, using geospatial methods. The results shows that around 190 million West Africans lack access to electricity. The first scenarios based on exsisting grid indicates 17% mini-grids, 43% grid extension and 40% solar home systems (SHSs). Whereas the planned grid scenario outcome indicates 12% mini-grids, 58% grid extension and 30% SHSs, hence more by grid extensions. Another research on electrification planning in SSA by 2030 [62], shows an investment need between a low of 19.3 bUSD (20% grid expansion, 0% mini-grid and 80% SHS) and a high of 370.6 bUSD (78% grid expansion, 16% mini-grid and 6% SHS) to provide universal access in WA [62]. The grid extension approach appears to dominate in all electrification expansion scenarios [61,62]. However, this approach may not eradicate energy poverty in regions like WA, due to complexity of rural settlements, poorly developed infrastructure and costs of grid extension [63]. Thus, off-grid solutions like mini-grids and especially SHS provide a least cost and potentially very fast applicable solutions for achieving rural electrification [63,64]. A recent study analysed the complete implications of mini-grid/off-grid electrification plans with a solar PV plus battery storage technology option for Nigeria [65]. According to Ref. [65], a 233 cluster covering 7.2 million people requiring 3280 MW solar PV and 7518 MWh battery capacities was derived as the best location for mini-grids auctions in Nigeria, while LCOE ranged from low 0.18 USD/kWh in the North to a high 0.25 USD/kWh in Southern Nigeria.

4.5. Benefits of a 100% RE system

The modelling outcomes show that a 100% RE system is the least-cost, least-GHG-emitting and most job-rich option for the WA power system, as observed in the BPSs. The PV-battery hybrid system appears as the backbone of the least-cost generation mix by 2050. The LCOE obtained for the BPSs is comparable to Barasa et al. [26], which shows a range of 35.2–47.6 €/MWh for SSA. The annualised cost of the system obtained for the year 2050 is 23.9 b€, 23.8 b€, 27.2 b€, 26.1 b€, 78.9 b€ and 47.5 b€ in the BPS-A, BPSnoCC-A, BPS-C, BPSnoCC-C, CPS-A and CPSnoCC-A, respectively. With GHG emissions costs, the total annualised cost is 70% higher in CPS than in the BPS, and is 50% higher without GHG emissions costs. The cost reduction effect observed in this study matches the experience of the energy transition in Uruguay. Uruguay made dramatic shifts to about 95% electricity from RE in the past 10 years, while reducing costs, building on its existing hydropower and new RE mainly wind energy, with some solar PV and bioenergy [66]. The results of this research show that the RE-based system is not only technically feasible, but also economically viable in WA. Such a finding is characteristic for 100% RE systems all around the world as highlighted by Brown et al. [67]. Most of the cost reductions in the BPSs can be attributed to low costs of RE technologies particularly solar PV and batteries. The costs of investments in RE technologies in the BPSs are offset by savings in fuel costs from displaced fossil-based generators, which is similar to the finding of Adeoye and Spataru [68], for WA region based on overnight approach for the year 2030. However, investments in fossil-based generators, as observed in the CPSs, will continue to be a burden on the WA's economy. The high costs in the CPSs are due to new investments in thermal generators, which are finally run on high full load hours that lead to enormous fuel costs. Furthermore, investments in fossil technologies are prone to cost overruns and schedule spills [69], they violate all sustainable criteria discussed in Ref. [70] and are likely to become stranded assets [35].

Beyond the robust techno-economic analysis of WA power systems, the LUT model also computes the emissions trajectory under various scenarios. The emissions pattern depicts the trajectory of RE deployment under various scenarios. In the BPSs, zero emission is achieved by 2050, however, the BPS across the area shows a rapid decarbonisation. Consequently, GHG emissions in the CPSs increased to about 60 $MtCO_{2eq}$ in 2030, which is comparable to the findings in Refs. [30], which shows a range of 70–80 $MtCO_{2eq}$ in 2030. By 2050, GHG emissions in the CPSs are about 209 $MtCO_{2eq}$. Investigating the impact of GHG emissions costs throughout the transition, the BPSs achieved a rapid decline in GHG emission in comparison to applying no GHG emissions costs. Without GHG emissions costs, the RE electricity generation reaches 99.9% (722.2 TWh) in the BPS across the area by 2050, while the remaining 0.1% (0.5 TWh) is supplied by gas turbines using fossil gas. Conversely, the RE generation in the BPS across the countries reaches 98.4% (721.0 TWh), while the remaining 1.6% (11.5 TWh) is supplied by gas turbines using fossil gas.

The energy transition is expected to lead the creation of new jobs, transformation or substitution of existing jobs and elimination of certain jobs, either as a complete phase out or at least a significant reduction without direct replacement [25]. The RE development in WA will facilitate socio-economic development, particularly job creation. Employment creation under various scenarios is examined in this research. The BPSs will create more jobs in comparison to the CPSs. Opponents of renewable energy often question if the sector can ever realistically create the numbers of jobs as a system based on large-scale fossil generators [25]. The results of this study show that RE dominated scenarios create more than twice the jobs compared to fossil-based scenarios. A 100% RE system would employ 440 thousand people in WA and solar PV emerges as the major job creating industry, employing about 300 thousand in 2050. A study found that replacing the millions of kerosene lamps, candles and flashlights with modern solar lighting technologies for people living off-grid could create 500 thousand new light-related jobs in the ECOWAS region [71]. Several studies [25,34,71] indicate that a shift in the energy landscape from fossil to RE technologies creates more jobs, which is comparable to the findings of this research.

4.6. Impact of changing parameters on results

Our findings indicate the significance of PV technologies and batteries as vital for the transition, due to anticipated decline in CAPEX. However, it is worth mentioning that changing cost parameters would influence technology deployment, as it appears that the costs and share of installed capacities of technologies are directly related, which is a predominant aspect of cost optimal future energy systems. Latest insights of Vartiainen et al. [72] indicate that the applied PV and battery CAPEX assumptions are rather conservative.

It should be noted that a uniform WACC of 7% is applied across the region, which is in line with leading international reports such as of IEA [73], IRENA [24] and journal publications [74–76]. However, Egli et al. [77] argue that uniform cost of capital (CoC) may result in distorted results and policies. It is worth mentioning that there are no better methods on offer to estimate future CoC values, uniform CoC remains the most ideal solution available. According to Bogdanov et al. [78] “proper choice of CoC assumption has a strong impact on the results of any energy system analysis and its uncertainty has to be considered in an analysis of the results”. Further, not all countries in WA fulfil the WACC assumption today; some uncertainty is induced by the WACC assumption. However, economic development in WA will lead to a more stable and lower WACC during the transition.

5. Conclusions and policy implications

The modelling outcomes show two distinctive features for the WA power system: expansion and transition. This study foresees transition of the WA power sector in exploiting the region’s abundant RE resources, which includes solar PV, wind energy, biomass and hydropower. Solar PV emerges as the new bulk electricity provider of the WA power sector as illustrated in the BPSs. Solar PV dominates in all the BPSs by 2050, and contributes the most (87–89%) to the total installed capacities and (81–85%) overall generation mix, respectively. In WA a 100% RE-based power system can run with low seasonal storage based on PtG conversion as observed in the BPS across the countries, whereas seasonal PtG storage is not required in the BPS across the area. Dispatchable RE hydropower reservoirs and bioenergy maintain a vital functionality in balancing seasonal variations in the WA power system. The solar PV, wind energy, hydropower and bioenergy synergies lead to a substantial reduction in storage requirements. The risk of day-to-day variability and capacity requirements reduce with higher transmission interconnection among the sub-regions, leading to increased system flexibility, broad cost savings and rapid defossilisation.

This study shows the need for substantial power generation capacity expansion due to under-capacity and growing demand in WA. Electricity demand grows 10 times by 2050 in comparison to 2015, while the installed capacity ranges from about 325 to 370 GW in the BPSs, and is about 130 GW in the CPSs. The capacity requirements are about 3 times higher in the BPSs than in the CPSs. This is due to RE technologies running on low FLH, particularly solar PV, while total electricity generation in the BPSs is around 6% higher than in CPSs. What is more, there are on-going plans to construct new fossil-based power plants in WA. These power plants are susceptible to cost overruns and schedule spills. In addition, the discoveries of new fossil fuels might pose a challenge to RE development in the region. However, this research presents a critical dimension of the transition for WA, which has to do with the cost-competitiveness of RE technologies as observed in the BPSs in comparison to the CPSs. Aside, the monumental and mounting costs of fossil-based technologies in the CPSs, such power systems violate sustainability principles that form the basis of a resilient power system. What is more, the BPSs without GHG emissions costs lead to 99.9% renewable in the BPS across the area and 98% in the BPS across the countries, while the remaining generation is covered by fossil fuel-fired gas turbines. This indicates pure market economics, even if the massive cost for GHG emissions cost would be allocated fully as subsidies for the fossil system by neglecting any price for the massive climate change costs. It will be imprudent for WA countries to invest in new fossil technologies, particularly coal power plants, but also nuclear. The fleets of RE technologies are accompanied by flexible gas turbines in the BPSs, however with fast declining full load hours. No new coal and nuclear power plant capacities are built in the least-cost expansion pathways for WA.

The ECOWAS energy policy should place solar PV at its core. Hybrid solar PV-battery power systems appear the least-cost solution for the region. A strong transmission grid infrastructure can enable substantial wind electricity generation from Niger and Mali, which can further reduce the entire energy system LCOE of WA. This research shows that a fully renewable electricity system is technically feasible and economically viable. A 100% RE system is a most efficient, least-cost, least GHG emission and most job-rich option for WA and is compatible with the Paris Agreement and the Sustainable Development Goals of the United Nations. This kind of power system is achievable with the vast RE resources of WA and it represents a real policy option for the region. It is noteworthy that a substantial roll-out of RE capacities in the BPSs does not require any subsidies at all. It is a least-cost option for the WA power system without subsidies. Consequently, subsidies might be unavoidable if a policy-driven deviation from this techno-economic least-cost option is pursued. A well-designed policy framework that limit new investments in conventional power plants, comprehensive energy market reforms and ambitious RE targets with a long-term perspective is needful in WA. Additionally, supportive policies that will strengthen cross-border electricity trade are vital for the WAPP. Further research has to be carried out incorporating additional energy sectors, i.e. heat, transport, industry and desalination, for broader analyses and to better understand the benefits of sector coupling for developing regions like WA.

Credit author statement

Ayobami Solomon Oyewo carried out main parts of the research and writing the manuscript. Arman Aghahosseini contributed to the research development. Manish Ram contributed to the research development. Christian Breyer framed the research questions and scope of the work, checked the results, facilitated discussions, and reviewed the manuscript.

Declaration of competing interest

None.

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Appendix A. Supplementary data

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